

HSPT Exam Simulator

Full-Length Practice Exam 07

Complete printable test with 298 multiple-choice questions, answer key, and detailed explanations.

Section	Questions	Time
Verbal Skills	60	20 minutes
Quantitative Skills	52	35 minutes
Reading Comprehension	62	35 minutes
Mathematics	64	45 minutes
Language	60	30 minutes

General Directions

Work through each section within the recommended time. Choose the best answer for every question. Mark one answer (A-D) for each item. Use the answer key and explanations only after completing the test.

Recommended total testing time: 165 minutes. Calculators should be used only when permitted by the rules followed for your HSPT preparation.

Section 1: Verbal Skills

60 questions | Recommended time: 20 minutes

1. Which word is closest in meaning to resilient?

- A) distant
- B) fragile
- C) ordinary
- D) adaptable

2. Which word is closest in meaning to meticulous?

- A) hurried
- B) careful
- C) uncertain
- D) cheerful

3. Which word is closest in meaning to abundant?

- A) plentiful
- B) costly
- C) hidden
- D) narrow

4. Which word is most nearly opposite in meaning to rigid?

- A) rough
- B) flexible
- C) plain
- D) heavy

5. Which word is most nearly opposite in meaning to scarce?

- A) useful
- B) divided
- C) abundant
- D) quiet

6. Which word is most nearly opposite in meaning to conceal?

- A) gather
- B) protect
- C) carry
- D) reveal

7. Sapling is to tree as colt is to ____.

- A) stable
- B) pasture
- C) horse
- D) saddle

8. Clay is to pottery as wool is to ____.

- A) loom
- B) sheep
- C) thread
- D) fabric

9. Compass is to direction as thermometer is to ____.

- A) weather
- B) measurement
- C) glass
- D) temperature

10. Which word does not belong with the others?

- A) sonnet
- B) novel
- C) haiku
- D) limerick

11. Copper, iron, and aluminum belong to the same group. Which word belongs with them?

- A) silver
- B) glass
- C) oak
- D) granite

12. Which word does not belong with the others?

- A) vineyard
- B) orchard
- C) aquarium
- D) plantation

13. Four presentations—J, K, L, and M—are scheduled one at a time. K must occur after J but before M. L cannot be first. Which schedule could be valid?

- A) J, K, L, M
- B) J, M, K, L
- C) K, J, M, L
- D) L, J, K, M

14. Every cedar in a certain grove is evergreen. Tree T is a cedar in that grove. Which statement must be true?

- A) Tree T is the tallest tree.
- B) Tree T has cones this year.
- C) Tree T is older than the other trees.
- D) Tree T is evergreen.

15. A display uses exactly three shelves. The blue vase must be above the amber vase. The green vase must be below the amber vase. Which arrangement is required from top to bottom?

- A) amber, blue, green
- B) blue, amber, green
- C) blue, green, amber
- D) green, amber, blue

16. Which word is closest in meaning to prudent?

- A) reckless
- B) cautious
- C) noisy
- D) brief

17. Which word is closest in meaning to diminish?

- A) reduce
- B) predict
- C) decorate
- D) combine

18. Which word is closest in meaning to impartial?

- A) impatient
- B) unbiased
- C) unusual
- D) uncertain

19. Which word is most nearly opposite in meaning to intricate?

- A) costly
- B) simple
- C) ancient
- D) polished

20. Which word is most nearly opposite in meaning to reluctant?

- A) silent
- B) formal
- C) patient
- D) eager

21. Which word is most nearly opposite in meaning to expand?

- A) contract
- B) observe
- C) settle
- D) arrange

22. Key is to lock as password is to ____.

- A) account
- B) message
- C) screen
- D) keyboard

23. Bee is to hive as rabbit is to ____.

- A) field
- B) carrot
- C) burrow
- D) fur

24. Editor is to manuscript as mechanic is to ____.

- A) garage
- B) vehicle
- C) wrench
- D) customer

25. Which word does not belong with the others?

- A) viola
- B) violin
- C) cello
- D) flute

26. Tulip, daffodil, and crocus belong to the same group. Which word belongs with them?

- A) fern
- B) rose
- C) moss
- D) hyacinth

27. Which word does not belong with the others?

- A) ruby
- B) emerald
- C) copper
- D) sapphire

28. If the museum opens late, the first tour is canceled. The first tour was not canceled. What must be true?

- A) The museum closed early.
- B) A second tour was canceled.
- C) The museum did not open late.
- D) The museum had no visitors.

29. Three boxes—P, Q, and R—are stacked. P is not on top. Q is below R. Which box must be on top?

- A) R
- B) P
- C) Q
- D) It cannot be determined.

30. Every member of the robotics team attends the safety meeting. Some students attending the safety meeting are eighth graders. Which conclusion is justified?

- A) Every eighth grader is on the robotics team.
- B) None of the first three conclusions must be true.
- C) No robotics team member is an eighth grader.
- D) Some robotics team members are eighth graders.

31. Which word is closest in meaning to scarce?

- A) limited
- B) ordinary
- C) plentiful
- D) hidden

32. Which word is closest in meaning to verify?

- A) confirm
- B) delay
- C) separate
- D) imagine

33. Which word is closest in meaning to resilient?

- A) fragile
- B) flexible
- C) able to recover
- D) careless

34. Which word is most nearly opposite in meaning to rigid?

- A) solid
- B) narrow
- C) careful
- D) flexible

35. Which word is most nearly opposite in meaning to conceal?

- A) protect
- B) measure
- C) reveal
- D) remove

36. Which word is most nearly opposite in meaning to abundant?

- A) useful
- B) scarce
- C) uneven
- D) rare

37. Archive is to records as pantry is to ____.

- A) shelves
- B) kitchen
- C) food
- D) recipe

38. Flour is to bread as clay is to ____.

- A) soil
- B) brick
- C) wheel
- D) water

39. Alarm is to warning as applause is to ____.

- A) approval
- B) performance
- C) audience
- D) sound

40. Which word does not belong with the others?

- A) daisy
- B) maple
- C) oak
- D) cedar

41. Quartz, granite, and marble belong to the same broad group. Which word belongs with them?

- A) cotton
- B) limestone
- C) glass
- D) leather

42. Which word does not belong with the others?

- A) stroll
- B) saunter
- C) sprint
- D) amble

43. Four students—J, K, L, and M—stand in a line. J is immediately before K. M is after K. L is before J. Who must be first?

- A) M
- B) J
- C) K
- D) L

44. A workshop offers pottery on Monday, drawing on Tuesday, and weaving on Wednesday. Nia can attend only after Monday and cannot attend drawing. Which workshop can she attend?

- A) weaving
- B) drawing
- C) none
- D) pottery

45. All cedar boxes in a storeroom are labeled. No labeled box is placed on the bottom shelf. Which statement must be true?

- A) Every labeled box is cedar.
- B) Some cedar boxes are on the top shelf.
- C) No cedar box is on the bottom shelf.
- D) Every bottom-shelf box is unlabeled wood.

46. Which word is closest in meaning to candid?

- A) secretive
- B) awkward
- C) lengthy
- D) honest

47. Which word is closest in meaning to feasible?

- A) expensive
- B) famous
- C) temporary
- D) possible

48. Which word is closest in meaning to vigilant?

- A) watchful
- B) confused
- C) sleepy
- D) generous

49. Which word is most nearly opposite in meaning to tranquil?

- A) peaceful
- B) distant
- C) restless
- D) orderly

50. Which word is most nearly opposite in meaning to permanent?

- A) lasting
- B) visible
- C) temporary
- D) useful

51. Which word is most nearly opposite in meaning to precise?

- A) exact
- B) formal
- C) complete
- D) vague

52. Architect is to blueprint as composer is to ____.

- A) audience
- B) concert
- C) instrument
- D) score

53. Chapter is to book as scene is to ____.

- A) actor
- B) stage
- C) play
- D) costume

54. Spark is to fire as seed is to ____.

- A) soil
- B) plant
- C) water
- D) garden

55. Which word does not belong with the others?

- A) mercury
- B) comet
- C) mars
- D) venus

56. Rectangle, rhombus, and trapezoid belong to the same broad group. Which word belongs with them?

- A) circle
- B) triangle
- C) pentagon
- D) kite

57. Which word does not belong with the others?

- A) expert
- B) beginner
- C) novice
- D) apprentice

58. A display has four shelves: top, upper-middle, lower-middle, and bottom. A globe is above a clock. A vase is below the clock. A model ship is above the globe. Which object must be on the bottom shelf?

- A) vase
- B) globe
- C) clock
- D) model ship

59. Every student in the robotics club attends either the Tuesday session or the Thursday session, but not both. Priya is in the robotics club and does not attend Tuesday. What must be true?

- A) Priya attends both sessions.
- B) Priya attends Thursday.
- C) Priya is not in the club.
- D) Priya attends neither session.

60. Three folders—red, blue, and green—are placed in three consecutive slots. The red folder is not first. The green folder is immediately before the red folder. Which folder must be first?

- A) blue
- B) red
- C) green
- D) It cannot be determined.

Section 2: Quantitative Skills

52 questions | Recommended time: 35 minutes

61. What number comes next? 7, 11, 18, 28, 41, ____

- A) 55
- B) 57
- C) 54
- D) 58

62. What number comes next? 96, 48, 52, 26, 30, ____

- A) 13
- B) 17
- C) 15
- D) 19

63. What number comes next? 2, 5, 4, 10, 6, 15, 8, ____

- A) 18
- B) 20
- C) 21
- D) 24

64. What number comes next? $\frac{1}{2}$, $\frac{3}{4}$, 1, $1\frac{1}{4}$, ____

- A) $1\frac{1}{3}$
- B) $1\frac{1}{2}$
- C) $1\frac{3}{8}$
- D) $1\frac{3}{4}$

65. A number is increased by 25% and becomes 75. What was the original number?

- A) 62.5
- B) 65
- C) 60
- D) 70

66. Which value is equivalent to 0.375?

- A) $\frac{3}{8}$
- B) $\frac{5}{16}$
- C) $\frac{7}{20}$
- D) $\frac{2}{5}$

67. A recipe uses flour and oats in a ratio of 5:2. If 15 cups of flour are used, how many cups of oats are needed?

- A) 6
- B) 7.5
- C) 10
- D) 5

68. Quantity A: $\frac{7}{12}$ of 84

Quantity B: $\frac{5}{8}$ of 80

- A) Quantity A is greater.
- B) The relationship cannot be determined from the information given.
- C) The two quantities are equal.
- D) Quantity B is greater.

69. Quantity A: y , where $\frac{y}{3} + 5 = 14$

Quantity B: 26

- A) The two quantities are equal.
- B) Quantity B is greater.
- C) Quantity A is greater.
- D) The relationship cannot be determined from the information given.

70. Quantity A: The greatest possible value of n if n is an integer and $n < 12$

Quantity B: 11

- A) The two quantities are equal.
- B) Quantity B is greater.
- C) Quantity A is greater.
- D) The relationship cannot be determined from the information given.

71. Quantity A: The area of a rectangle with length 9 cm and width 6 cm

Quantity B: The area of a triangle with base 12 cm and height 9 cm

- A) Quantity A is greater.
- B) Quantity B is greater.
- C) The relationship cannot be determined from the information given.
- D) The two quantities are equal.

72. Quantity A: The perimeter of a regular pentagon with side length 7 cm

Quantity B: 36 cm

- A) Quantity B is greater.
- B) Quantity A is greater.
- C) The two quantities are equal.
- D) The relationship cannot be determined from the information given.

73. Quantity A: The volume of a cube with edge length 4 cm

Quantity B: The volume of a rectangular prism measuring 8 cm by 4 cm by 2 cm

- A) Quantity A is greater.
- B) The two quantities are equal.
- C) Quantity B is greater.
- D) The relationship cannot be determined from the information given.

74. What number comes next? 81, 72, 64, 57, 51, ____

- A) 45
- B) 44
- C) 46
- D) 43

75. What number comes next? 3, 6, 18, 72, 360, ____

- A) 2,160
- B) 1,440
- C) 1,800
- D) 2,520

76. What number comes next? 90, 82, 67, 45, 16, ____

- A) -13
- B) -18
- C) -20
- D) -22

77. Which number is divisible by both 6 and 9?

- A) 42
- B) 84
- C) 72
- D) 54

78. If $\frac{3}{5}$ of a number is 27, what is the number?

- A) 40
- B) 45
- C) 42
- D) 48

79. A price decreases from \$80 to \$68. What is the percent decrease?

- A) 20%
- B) 12%
- C) 18%
- D) 15%

80. Which expression has the greatest value?

- A) 2.4×5
- B) $18 \div 1.5$
- C) $\frac{3}{4}$ of 20
- D) 40% of 35

81. Quantity A: 35% of 240

Quantity B: $\frac{7}{10}$ of 120

- A) Quantity A is greater.
- B) Quantity B is greater.
- C) The two quantities are equal.
- D) The relationship cannot be determined from the information given.

82. Quantity A: x , if $4x - 7 = 29$

Quantity B: 10

- A) Quantity A is greater.
- B) Quantity B is greater.
- C) The two quantities are equal.
- D) The relationship cannot be determined from the information given.

83. Quantity A: The least possible integer k if $k > -3$

Quantity B: -2

- A) Quantity A is greater.
- B) Quantity B is greater.
- C) The relationship cannot be determined from the information given.
- D) The two quantities are equal.

84. Quantity A: The area of a square with perimeter 32 cm

Quantity B: 60 cm^2

- A) Quantity A is greater.
- B) Quantity B is greater.
- C) The two quantities are equal.
- D) The relationship cannot be determined from the information given.

85. Quantity A: The diameter of a circle with radius 6 cm

Quantity B: The side length of an equilateral triangle with perimeter 39 cm

- A) Quantity A is greater.
- B) The two quantities are equal.
- C) Quantity B is greater.
- D) The relationship cannot be determined from the information given.

86. Quantity A: The area of a rhombus with diagonals 10 cm and 12 cm

Quantity B: The area of a triangle with base 15 cm and height 7 cm

- A) Quantity A is greater.
- B) Quantity B is greater.
- C) The two quantities are equal.
- D) The relationship cannot be determined from the information given.

87. What number comes next? 2, 5, 11, 23, 47, ____

- A) 71
- B) 83
- C) 97
- D) 95

88. What number comes next? 100, 96, 88, 76, 60, ____

- A) 42
- B) 44
- C) 40
- D) 36

89. What number comes next? 1, 4, 2, 8, 4, 16, ____

- A) 6
- B) 12
- C) 32
- D) 8

90. What number comes next? 0.5, 1, 0.8, 1.6, 1.4, 2.8, ____

- A) 2.4
- B) 3.0
- C) 2.6
- D) 5.6

91. A recipe uses 2 cups of oats for every 3 cups of nuts. If 12 cups of nuts are used, how many cups of oats are needed?

- A) 6
- B) 8
- C) 9
- D) 18

92. Which value is equivalent to 0.375?

- A) $\frac{3}{5}$
- B) $\frac{5}{8}$
- C) $\frac{7}{20}$
- D) $\frac{3}{8}$

93. A number is increased by 25% and becomes 90. What was the original number?

- A) 68
- B) 70
- C) 72
- D) 75

94. Quantity A: $|-11| - 3$

Quantity B: 2^3

- A) Quantity A is greater.
- B) The two quantities are equal.
- C) Quantity B is greater.
- D) The relationship cannot be determined from the information given.

95. Quantity A: y , if $y/4 + 2 = 7$

Quantity B: 18

- A) The relationship cannot be determined from the information given.
- B) Quantity B is greater.
- C) The two quantities are equal.
- D) Quantity A is greater.

96. Quantity A: $m + 5$

Quantity B: 12

Given only that m is a positive integer less than 10.

- A) Quantity A is greater.
- B) Quantity B is greater.
- C) The relationship cannot be determined from the information given.
- D) The two quantities are equal.

97. Quantity A: The perimeter of a regular pentagon with side length 7 cm

Quantity B: 36 cm

- A) Quantity A is greater.
- B) Quantity B is greater.
- C) The two quantities are equal.
- D) The relationship cannot be determined from the information given.

98. Quantity A: The volume of a rectangular prism measuring 3 cm by 4 cm by 5 cm

Quantity B: 60 cm^3

- A) The two quantities are equal.
- B) Quantity B is greater.
- C) Quantity A is greater.
- D) The relationship cannot be determined from the information given.

99. Quantity A: The measure of an exterior angle supplementary to a 112° interior angle

Quantity B: 70°

- A) Quantity A is greater.
- B) Quantity B is greater.
- C) The two quantities are equal.
- D) The relationship cannot be determined from the information given.

100. What number comes next? 8, 14, 26, 50, 98, ____

- A) 194
- B) 180
- C) 196
- D) 198

101. What number comes next? 162, 54, 18, 6, ____

- A) 2
- B) 1
- C) 3
- D) 4

102. What number comes next? 4, 9, 8, 13, 12, 17, ____

- A) 14
- B) 16
- C) 18
- D) 22

103. A value is multiplied by 0.4 and the result is 18. What is the original value?

- A) 7.2
- B) 45
- C) 54
- D) 72

104. Which number leaves a remainder of 2 when divided by 5?

- A) 35
- B) 50
- C) 41
- D) 47

105. If $\frac{5}{6}$ of a quantity is 35, what is $\frac{2}{7}$ of the quantity?

- A) 12
- B) 14
- C) 18
- D) 10

106. A price is reduced from \$80 to \$68. What percent decrease is this?

- A) 18%
- B) 12%
- C) 15%
- D) 20%

107. Quantity A: $\frac{3}{5}$ of 45

Quantity B: 60% of 45

- A) Quantity A is greater.
- B) Quantity B is greater.
- C) The relationship cannot be determined from the information given.
- D) The two quantities are equal.

108. Quantity A: n^2

Quantity B: n

Given that $-1 < n < 0$.

- A) Quantity A is greater.
- B) Quantity B is greater.
- C) The two quantities are equal.
- D) The relationship cannot be determined from the information given.

109. Quantity A: $p + q$

Quantity B: 12

Given that $p - q = 4$.

- A) The relationship cannot be determined from the information given.
- B) Quantity B is greater.
- C) The two quantities are equal.
- D) Quantity A is greater.

110. Quantity A: The area of a triangle with base 12 cm and height 5 cm

Quantity B: 32 cm^2

- A) Quantity A is greater.
- B) The relationship cannot be determined from the information given.
- C) The two quantities are equal.
- D) Quantity B is greater.

111. Quantity A: The diameter of a circle with radius 9 cm

Quantity B: 18 cm

- A) Quantity A is greater.
- B) Quantity B is greater.
- C) The relationship cannot be determined from the information given.
- D) The two quantities are equal.

112. Quantity A: The third side of a triangle whose other sides are 6 cm and 10 cm

Quantity B: 8 cm

- A) Quantity A is greater.
- B) The relationship cannot be determined from the information given.
- C) The two quantities are equal.
- D) Quantity B is greater.

Section 3: Reading Comprehension

62 questions | Recommended time: 35 minutes

Literary Passage: The Last Rehearsal

Nia had volunteered to operate the sound board for the school play because she preferred working behind the curtain. During rehearsals, she learned which microphone belonged to each actor and marked every adjustment in a notebook.

On the afternoon of the final rehearsal, the drama teacher announced that the usual sound operator would be absent on opening night. "You already know the board better than anyone else here," she told Nia. "Can you run it tomorrow?"

Nia's first impulse was to refuse. Practicing with an empty auditorium felt different from controlling sound while families filled every seat. Still, she agreed to try the entire rehearsal as though it were opening night.

Halfway through the second scene, a microphone suddenly produced a sharp burst of noise. Nia lowered its channel, switched the actor to a backup microphone, and wrote down what had happened. She expected the teacher to be disappointed. Instead, the teacher nodded and continued the scene.

After rehearsal, Nia checked the cable and discovered a loose connection. The problem had not ruined the practice; it had shown her that she could respond without freezing.

As she closed her notebook, Nia still felt nervous about opening night. Yet the nervousness no longer seemed like evidence that she was unprepared. It felt more like a signal to pay attention.

113. Why had Nia marked adjustments in a notebook?

- A) To record the actors' lines
- B) To keep track of sound-board settings
- C) To list audience members
- D) To write a review of the play

114. Why does the teacher most likely continue the scene after the microphone problem?

- A) She sees that Nia has handled the problem effectively.
- B) She does not notice the noise.
- C) She wants the actor to stop using microphones.
- D) She believes the rehearsal is unimportant.

115. In the final paragraph, "signal" most nearly means

- A) an indication or reminder.
- B) a command from the teacher.
- C) a sound produced by equipment.
- D) a written schedule.

116. How does Nia's view of nervousness change?

- A) She decides nervousness always causes mistakes.
- B) She comes to see nervousness as something that can sharpen her attention.
- C) She believes nervousness proves she should quit.
- D) She stops feeling nervous entirely.

117. Which statement best expresses a central idea of the passage?

- A) Technical jobs are easier than performing onstage.
- B) Confidence can grow from successfully responding to an unexpected problem.
- C) Rehearsals should never include equipment failures.
- D) Careful notes can prevent every mistake.

Informational Passage: Why Some Cities Build Rain Gardens

During a heavy storm, water runs off roofs, sidewalks, parking lots, and streets. In areas covered by many hard surfaces, much of that water cannot soak directly into the ground. It may instead flow rapidly into storm drains, sometimes carrying soil, litter, and other materials with it.

A rain garden is a shallow planted area designed to temporarily collect some of this runoff. It is usually placed where water naturally flows from a roof, driveway, or paved area. The garden's soil and plants slow the water and allow part of it to soak into the ground.

Rain gardens are not simply ordinary flower beds placed in wet locations. Designers consider the amount of runoff, how quickly the soil drains, and which plants can tolerate both wet and dry periods. A garden that holds water too long may not function as intended, while one that drains too quickly may capture little runoff.

A single rain garden cannot solve every storm-water problem in a neighborhood. However, many small features can work together with drains, ponds, and other systems. They can also add vegetation to developed areas.

For planners, the value of a rain garden lies partly in its scale. It is a relatively small feature, but when carefully located and designed, it can help manage water close to where the rain falls.

118. According to the passage, what is one function of a rain garden?

- A) To send runoff more quickly into streets
- B) To replace every storm drain in a neighborhood
- C) To keep all planted areas permanently wet
- D) To slow runoff and allow some water to soak into the ground

119. As used in the first paragraph, “rapidly” most nearly means

- A) quietly.
- B) completely.
- C) quickly.
- D) rarely.

120. Which rain garden would most likely be poorly designed?

- A) One that remains filled with water far longer than intended
- B) One placed where runoff naturally reaches it
- C) One sized after the amount of runoff is considered
- D) One whose plants can tolerate changing moisture

121. How is the passage mainly organized?

- A) It describes several storms from weakest to strongest.
- B) It compares two cities and argues that one has better parks.
- C) It gives chronological instructions for planting a home garden.
- D) It presents a runoff problem, explains one response, and discusses the response's limits and design needs.

122. Which factor do designers consider when planning a rain garden?

- A) The color of nearby buildings
- B) The age of the nearest street
- C) How quickly the soil drains
- D) The number of windows on nearby houses

Humanities Passage: Reading an Old Map

An old map can look authoritative. Its lines are precise, its place names are carefully written, and its symbols seem to turn a complicated landscape into an orderly page. Yet historians do not treat a map as a perfectly neutral picture of the world.

Every map reflects choices. A mapmaker decides what area to include, what to leave out, which features deserve labels, and how large or prominent those features should appear. A harbor map may emphasize water depth and channels, while a political map may highlight borders. Neither map is necessarily inaccurate simply because it focuses on different information.

The map's intended audience also matters. A document prepared for merchants may identify roads, ports, and market towns. A military map from the same period may devote more attention to hills, bridges, and defensive positions. Comparing the two can reveal not only geographic information but also the priorities of the people who made and used them.

Historians therefore ask questions beyond "Where was this place?" They ask who produced the map, for whom, and for what purpose. They may compare it with letters, surveys, or later maps.

Seen this way, a map is both a source of geographic evidence and a record of human decisions. Its usefulness increases when readers examine what it shows, what it omits, and why those choices may have been made.

123. What is the main idea of the passage?

- A) Old maps are generally too inaccurate to be useful.
- B) Maps provide evidence that should be interpreted in light of the mapmaker's choices and purpose.
- C) Historians value maps only for locating roads and borders.
- D) Political maps are more useful than other kinds of maps.

124. In the first paragraph, "authoritative" most nearly means

- A) old-fashioned.
- B) reliable and convincing.
- C) difficult to read.
- D) colorful.

125. What can reasonably be inferred from the comparison of merchant and military maps?

- A) Military maps never included towns.
- B) Merchants were responsible for drawing all early maps.
- C) Different users may need different information about the same region.
- D) Maps made for different audiences cannot be compared.

126. Why does the author contrast a harbor map with a political map?

- A) To prove that harbor maps contain more detail
- B) To explain how maps are printed
- C) To show that political borders are usually inaccurate
- D) To illustrate how a map's purpose affects what it emphasizes

127. Which best describes the author's tone?

- A) humorous and casual
- B) analytical and measured
- C) suspicious and dismissive
- D) impatient and critical

128. Besides other maps, which sources does the passage say historians may use for comparison?

- A) novels and poems
- B) weather forecasts and photographs
- C) letters and surveys
- D) speeches and advertisements

Literary Passage: The Borrowed Lantern

Mara had agreed to help her neighbor Mr. Ito prepare a courtyard for the evening block gathering. Her job seemed simple: hang a row of paper lanterns beneath the covered walkway. Mr. Ito handed her a wooden box containing lanterns he had used for years.

Mara worked quickly until she noticed that one lantern had a narrow tear near its base. She considered placing it at the far end, where few people would notice. Instead, she carried it to Mr. Ito and apologized, assuming she had damaged it.

He examined the tear and smiled. "That happened last spring," he said. "I kept it because the frame is still good." Together they cut a piece of translucent paper and fitted it behind the torn section. The patch did not disappear. When the lantern was lit, it formed a darker shape across one side.

That evening, Mara watched visitors pass beneath the row. No one complained about the patched lantern. A small child even pointed to it and said the shadow looked like a mountain.

Mara had begun the afternoon thinking that a repaired object was merely a second-best version of a new one. By the time she left, she understood why Mr. Ito had kept the lantern: usefulness did not always require perfection.

129. Why does Mara first bring the torn lantern to Mr. Ito?

- A) She wants him to replace every lantern.
- B) She needs help reaching the walkway.
- C) She dislikes the lantern's color.
- D) She assumes she may have damaged it.

130. What does Mr. Ito's decision to keep the lantern suggest about him?

- A) He refuses to repair anything he owns.
- B) He prefers damaged objects to new ones.
- C) He values objects that can still serve a purpose.
- D) He does not notice imperfections.

131. What is the effect of the child's comment about the mountain-shaped shadow?

- A) It introduces a new conflict.
- B) It shows that an imperfection can be viewed creatively.
- C) It proves the repair is invisible.
- D) It explains why the lantern tore.

132. Which statement best expresses a central idea of the passage?

- A) Old objects should never be replaced.
- B) Community events require careful decoration.
- C) Something can remain worthwhile even when it is imperfect.
- D) Admitting mistakes always prevents problems.

133. As used in the fourth paragraph, “fitted” most nearly means

- A) measured.
- B) folded into a box.
- C) tested for brightness.
- D) placed so that it matched the space.

Scientific Passage: How Desert Beetles Collect Water

In some coastal deserts, rain is rare even though fog may arrive regularly. Certain beetles living in these dry regions can take advantage of moisture carried by fog. Their bodies do not create water; instead, their surfaces and behavior help them collect droplets already present in the air.

When fog moves across the dunes, a beetle may raise the rear of its body and face into the moving air. Tiny droplets gather on its outer surface. As droplets combine and become larger, gravity can help them move toward the beetle's mouth.

Scientists studying this process are interested in more than the beetles themselves. The insects demonstrate how shape, surface properties, and position can influence the movement of small amounts of water. Researchers have experimented with materials that imitate some of these features in hopes of improving devices that collect moisture from air.

Such designs do not mean that a beetle can simply be copied like a blueprint. A useful human-made collector must also account for local humidity, wind, cost, durability, and scale. Still, observing how organisms function in difficult environments can suggest principles that engineers might test in new ways.

134. What happens as tiny droplets on the beetle become larger?

- A) They evaporate immediately.
- B) They can move toward the beetle's mouth.
- C) They turn into salt.
- D) They cause the beetle to burrow.

135. What is the main purpose of the passage?

- A) To provide instructions for keeping beetles as pets
- B) To argue that deserts receive more rain than scientists report
- C) To compare the diets of several desert insects
- D) To explain a beetle's water-collection process and how it can inspire engineering research

136. Why does the author say a beetle cannot simply be copied like a blueprint?

- A) Beetles change shape every day.
- B) Scientists cannot observe beetles closely.
- C) Fog contains no usable water.
- D) Engineered collectors must satisfy conditions beyond the beetle's natural process.

137. How does the passage develop its ideas?

- A) It lists desert animals from smallest to largest.
- B) It presents a debate and chooses one side.
- C) It explains a natural process, then discusses a possible application and its limits.
- D) It gives step-by-step instructions for building a machine.

138. Which factor is specifically mentioned as important for a human-made water collector?

- A) Soil color
- B) Local humidity
- C) Number of insects nearby
- D) Length of daylight

Informational Passage: The Work Behind a Museum Label

A museum label may contain only a few sentences, but producing those sentences can require substantial research. Before a label is installed beside an object, museum staff must decide which facts are both reliable and useful to visitors.

The process often begins with records about the object's origin, materials, maker, or previous owners. Curators may compare these records with photographs, catalogs, letters, or scientific analysis. Sometimes the evidence is incomplete. In that case, a careful label distinguishes what is known from what is uncertain rather than presenting a guess as fact.

Writers also consider the visitor's experience. A label cannot include everything specialists know. If it becomes overloaded with dates and technical terms, readers may miss the central idea. Staff therefore select details that help explain why the object matters, while additional information may be provided through digital guides or longer publications.

Labels can also change. New research may correct an old identification, clarify an object's history, or provide a better interpretation of its use. Updating a label is not necessarily evidence that earlier staff were careless. It can show that historical knowledge develops as new evidence appears.

The small rectangle beside an exhibit is therefore the visible result of choices about evidence, accuracy, emphasis, and audience.

139. What should a careful label do when evidence is incomplete?

- A) Omit the object entirely.
- B) Distinguish known information from uncertainty.
- C) Present the most interesting guess as fact.
- D) Replace research with a visitor survey.

140. Which statement best summarizes the passage?

- A) Museum labels result from research and choices about accuracy, emphasis, and audience.
- B) Scientific analysis is the only reliable source for museum labels.
- C) Digital guides will eventually replace all museum labels.
- D) Museum labels are brief because objects have little history.

141. Why might staff leave a technical detail off a label even when it is accurate?

- A) They are forbidden to use technical information.
- B) They want the object's history to remain secret.
- C) They assume visitors never read labels.
- D) They may judge that it distracts from the most useful idea for visitors.

142. Why does the author call the label “the visible result” in the final paragraph?

- A) To emphasize that a short label reflects substantial work and decision-making
- B) To imply that museum objects are less important than labels
- C) To suggest that labels should be printed in brighter colors
- D) To show that visitors can see the staff writing labels

143. Which best describes the organization of the passage?

- A) It ranks museum departments by importance.
- B) It follows one curator through a single workday.
- C) It describes stages and considerations involved in creating and updating labels.
- D) It compares museums in different countries.

144. Why does the author discuss labels being updated?

- A) To argue that old labels should always be discarded
- B) To prove that museums frequently lose records
- C) To show that changing a label can reflect the growth of knowledge
- D) To explain how labels are physically attached

Literary Passage: The Unmarked Trail

When Eli joined his cousin Soren for a morning hike, he expected Soren to lead. Soren had walked the preserve many times and usually moved through it with the confidence of someone following a familiar hallway. Yet after a night of heavy wind, the path looked different. Pine needles covered the ground, and a fallen branch hid one of the wooden trail markers.

At a fork, Soren started toward the wider path. Eli noticed a shallow groove in the soil on the narrower route and remembered that the map showed the trail bending toward a creek before climbing the ridge. He pointed this out, but Soren hesitated. The wider path simply looked more convincing.

Rather than argue, Eli unfolded the map. They compared the creek's position with the slope beside them and chose the narrower route. Within ten minutes they reached the footbridge shown on the map.

Soren laughed. “I guess knowing a place isn't the same as checking where you are.”

On the return trip, they found the hidden marker beneath the fallen branch and propped it where it could be seen. Eli realized that confidence could be useful, but evidence was useful too—especially when familiar surroundings had changed.

145. What had hidden one of the trail markers?

- A) a fallen branch
- B) pine needles
- C) the creek
- D) a map

146. Why does Eli suggest the narrower route?

- A) He wants a more difficult hike.
- B) He sees the hidden marker from the fork.
- C) He knows the wider path is closed.
- D) He remembers map details that match the surrounding land.

147. Which idea is most central to the passage?

- A) Familiarity should always replace maps.
- B) Evidence can be especially valuable when familiar conditions change.
- C) Hiking is safest only with two people.
- D) Wide paths are usually misleading.

148. As used in the second paragraph, “convincing” most nearly means

- A) easy to repair
- B) physically dangerous
- C) unusually narrow
- D) likely to be correct

149. What does the hidden trail marker mainly symbolize in the story?

- A) Information that can become difficult to notice when circumstances change
- B) A competition between the cousins
- C) A reason to avoid nature preserves
- D) Soren's dislike of maps

Literary Passage: The Quiet Rehearsal

The community theater was empty except for Leena and Mr. Alvarez, who was repairing a loose hinge on a scenery panel. Leena had arrived early to practice a short piano introduction she would play before the evening performance.

During the first run-through, she rushed the final measures. No audience was present, yet she felt as if the empty seats were already judging her. She began again and made the same mistake.

Mr. Alvarez did not offer musical advice. Instead, he asked Leena to help him carry the repaired panel across the stage. They moved slowly because the painted surface could scrape against the floor.

When Leena returned to the piano, she noticed the deliberate pace of their walk still in her mind. She played the introduction again, this time allowing each phrase to finish before beginning the next. The final measures no longer felt like an obstacle she had to escape.

“Better?” Mr. Alvarez asked.

Leena nodded. She had assumed improvement would come from repeating the difficult notes faster and more often. Instead, a completely different task had reminded her to pay attention to pace.

150. Why do Leena and Mr. Alvarez carry the scenery panel slowly?

- A) Its painted surface could scrape against the floor.
- B) They are waiting for another actor.
- C) The stage lights are off.
- D) It is extremely heavy.

151. What most likely helps Leena improve her playing?

- A) She memorizes new notes.
- B) She decides the audience will not notice mistakes.
- C) Mr. Alvarez demonstrates the introduction.
- D) She connects the careful pace of carrying the panel with the pace of the music.

152. What does Leena learn?

- A) Practicing more quickly always improves accuracy.
- B) Musicians should avoid rehearsing in empty theaters.
- C) Stage work is more difficult than piano practice.
- D) An insight from one activity can change how a person approaches another.

153. Why does the author include the empty seats in the second paragraph?

- A) To show that Leena feels pressure even without an actual audience
- B) To explain why the theater is closing
- C) To show that the performance has been canceled
- D) To suggest that Mr. Alvarez dislikes crowds

154. How is the passage mainly organized?

- A) Several musicians debate the best practice method.
- B) A problem is introduced, an unrelated-seeming event occurs, and that event helps produce a solution.
- C) Events are presented in reverse chronological order.
- D) The author compares two theaters.

Informational Passage: Why Some Libraries Lend More Than Books

Many public libraries are best known for lending books, but some now lend objects such as basic tools, musical instruments, sewing equipment, or science kits. These collections are sometimes called “libraries of things.” The idea follows a familiar library principle: a useful resource does not always need to be individually owned. A person who needs a stud finder for one afternoon or a telescope for a weekend may prefer borrowing to purchasing. Shared collections can also give people a low-cost way to try an activity before deciding whether to invest in their own equipment.

Creating such a collection requires different planning from shelving books. Staff must consider storage, inspection, replacement parts, cleaning, and instructions for safe use. Some objects may be unsuitable because they are too fragile, expensive to maintain, or difficult to check between loans.

Libraries also study what their communities actually need. A collection useful in one neighborhood may be less valuable in another. For that reason, staff may survey patrons, track requests, and adjust the inventory over time. A library of things therefore extends the logic of lending without abandoning it. The objects are different, but the central question remains familiar: which shared resources can help a community gain practical access to information, skills, or useful experiences?

155. Which concern is specifically mentioned when libraries plan object-lending collections?

- A) The color of the objects
- B) The age of the library building
- C) Whether every object fits on a bookshelf
- D) Cleaning items between loans

156. What is the main idea of the passage?

- A) Libraries should stop lending books and lend tools instead.
- B) Every library should offer exactly the same objects.
- C) Libraries of things apply the principle of shared access to useful objects while requiring careful planning.
- D) Borrowing objects is always cheaper than buying them.

157. Why might two libraries choose very different objects for their collections?

- A) Only one library may lend each kind of object.
- B) Library rules forbid similar inventories.
- C) Community needs and requests can differ by location.
- D) Staff members choose objects randomly.

158. As used in the fourth paragraph, “inventory” most nearly means

- A) a written complaint
- B) a borrowing fee
- C) a storage room
- D) the collection of available items

159. Which best describes the structure of the passage?

- A) It tells the history of one library in chronological order.
- B) It presents two opposing arguments without reaching a conclusion.
- C) It defines a practice, explains benefits and challenges, then describes how local needs shape it.
- D) It gives instructions for repairing borrowed equipment.

Literary Passage: The Last Empty Frame

Nora volunteered to help arrange photographs for the neighborhood history exhibit. By late afternoon, nearly every frame on the long wall had been filled. One narrow frame remained empty between a photograph of the old train depot and a picture of the first public swimming pool.

Mr. Chen, the exhibit coordinator, handed Nora a box of extra photographs and asked her to choose one. She first selected a dramatic image of firefighters beside an engine. It was striking, but the caption showed that it had been taken in another town. She returned it to the box.

Next she found a faded photograph of several children carrying books across a muddy lot. The image seemed ordinary until Nora read the note on the back. The children were walking to the neighborhood's first temporary library, which had operated from two rented rooms before the current library was built.

Nora placed the photograph in the empty frame. It was less polished than many of the others, yet it connected two moments already represented on the wall: the neighborhood's growth and the services residents had built as they needed them.

When Mr. Chen returned, he studied Nora's choice and smiled. “Sometimes the quiet picture explains the most,” he said. Nora looked again at the children crossing the mud and understood what he meant.

160. Why does Nora reject the photograph of the firefighters?

- A) It is too large for the frame.
- B) Its caption is missing.
- C) It was taken in another town.
- D) Mr. Chen dislikes it.

161. Why does the faded photograph become more meaningful to Nora after she reads the note?

- A) She learns that it records an early stage in a local institution's history.
- B) She discovers one child is Mr. Chen.
- C) She realizes it is valuable because it is rare.
- D) She notices the lot is now a train depot.

162. What does the empty frame mainly help represent in the story?

- A) Nora's opportunity to decide what kind of history deserves attention
- B) Mr. Chen's poor planning
- C) The neighborhood's lack of important events
- D) A photograph that cannot be replaced

163. As used in the fourth paragraph, "polished" most nearly means

- A) shiny from cleaning
- B) professionally impressive in appearance
- C) recently printed
- D) carefully labeled

164. Which theme is best supported by the passage?

- A) Ordinary-looking evidence can reveal important parts of a community's story.
- B) Only dramatic events belong in exhibits.
- C) Old photographs are more accurate than written records.
- D) Exhibits should always be chronological.

Informational Passage: How Snow Fences Work

A snow fence does not stop snow from falling, and it is not designed to catch every flake that blows toward it. Instead, it changes the movement of wind near the ground.

When moving air reaches a properly designed snow fence, some air passes through the openings while some flows over the top. The wind slows on the downwind side. Because slower air can carry less blowing snow, snow particles settle out and form a drift at a predictable distance from the fence.

Placement matters. If a fence is installed too close to a road, the resulting drift may form where snow removal crews least want it. Engineers therefore consider typical wind direction, expected snow volume, fence height, and the distance between the fence and the area being protected.

Snow fences can be made from wooden slats, plastic mesh, or even rows of shrubs. Living rows of vegetation may provide additional benefits, such as wildlife habitat, but they take time to grow and require suitable land.

The key idea is counterintuitive: the fence protects a road or building not by creating a solid wall, but by deliberately causing snow to settle somewhere safer. Its success depends on guiding wind and snow rather than simply blocking them.

165. What happens to wind on the downwind side of a properly designed snow fence?

- A) It speeds up sharply.
- B) It slows down.
- C) It stops completely.
- D) It changes into warm air.

166. Why can placing a snow fence too close to a road create a problem?

- A) The fence may cause a drift where the road needs to remain clear.
- B) The fence prevents snowplows from starting.
- C) The road receives more snowfall from clouds.
- D) The fence always collapses into traffic.

167. As used in the final paragraph, “counterintuitive” most nearly means

- A) difficult to measure
- B) opposite to what one might first expect
- C) harmful to the environment
- D) based on an old method

168. How is the passage mainly organized?

- A) A mechanism, placement factors, examples, and a concluding principle
- B) A historical timeline
- C) Two competing scientific theories
- D) Instructions for one exact fence

169. What is the main purpose of the passage?

- A) To argue all fences should be shrubs
- B) To explain how snow fences use changes in wind speed to control where drifting snow settles
- C) To show why roads should close in winter
- D) To describe snowfall prediction

Humanities Passage: A Mapmaker's Margin Notes

In the early days of detailed city mapping, a finished map could represent months of walking, measuring, comparing records, and redrawing lines. Yet some of the most revealing parts of a mapmaker's work never appeared in the printed map. They remained in field notebooks and margin notes.

A surveyor might record that a lane had recently changed names, that a bridge was being rebuilt, or that residents used a shortcut absent from official records. Such notes helped explain why two sources disagreed. They also reminded the mapmaker which details were uncertain and needed another visit.

Modern researchers sometimes value these working notes for a different reason. A printed map presents decisions as settled: one name is chosen, one boundary is drawn, one road is included. Notes can reveal the questions behind those decisions. They show that maps are constructed from evidence that may be incomplete, changing, or disputed.

This does not make maps unreliable. Rather, it helps readers understand mapping as a process of judgment as well as measurement. A careful mapmaker does not merely collect facts; the mapmaker evaluates sources, notices changes, and decides how best to represent a place at a particular moment.

The margins, then, can preserve something the clean final map cannot: a record of uncertainty being examined rather than hidden.

170. According to the passage, what might a mapmaker record in field notes?

- A) A recently changed street name
- B) Only measurements printed on the final map
- C) The selling price of every building
- D) Personal opinions about travelers

171. Why might researchers study margin notes in addition to a finished map?

- A) The notes always contain more accurate distances.
- B) The notes are easier to read.
- C) The notes reveal uncertainties and decisions hidden by the map's finished appearance.
- D) The notes prove official records are false.

172. As used in the third paragraph, “settled” most nearly means

- A) occupied by residents
- B) placed at the bottom
- C) resolved or decided
- D) paid in full

173. What contrast does the author use most strongly?

- A) Old cities versus new cities
- B) Walking versus measuring
- C) Bridges versus roads
- D) Clean final maps versus working notes that preserve uncertainty

174. Which sentence best summarizes the passage?

- A) Working notes can reveal the judgments and uncertainties behind a finished map.
- B) Mapmakers once depended entirely on memory.
- C) Modern maps no longer require evidence.
- D) City boundaries are impossible to represent.

Section 4: Mathematics

64 questions | Recommended time: 45 minutes

175. What is the value of $18 - 3 \times 4 + 5$?

- A) 11
- B) 65
- C) 17
- D) 35

176. Which fraction is equivalent to 0.45?

- A) $\frac{4}{5}$
- B) $\frac{9}{20}$
- C) $\frac{5}{9}$
- D) $\frac{11}{20}$

177. What is $3^3 + \sqrt{49}$?

- A) 20
- B) 28
- C) 32
- D) 34

178. Which number has exactly three positive factors?

- A) 49
- B) 45
- C) 51
- D) 57

179. If $3(x + 4) = 27$, what is x ?

- A) 13
- B) 7
- C) 9
- D) 5

180. Which expression represents “six less than twice a number n ”?

- A) $6 - 2n$
- B) $2(n - 6)$
- C) $2n - 6$
- D) $6n - 2$

181. A taxi fare consists of a \$4 starting charge plus \$2.50 per mile. Which expression gives the fare F for m miles?

- A) $F = 4 + 2.50m$
- B) $F = 6.50m$
- C) $F = 4m + 2.50$
- D) $F = 2.50 + 4m$

182. A number is multiplied by 5, and then 8 is subtracted. The result is 32. What is the number?

- A) 8
- B) 7
- C) 10
- D) 6

183. The angles of a quadrilateral measure 88° , 92° , 105° , and x° . What is x ?

- A) 95°
- B) 65°
- C) 85°
- D) 75°

184. A rectangular prism is 5 cm long, 3 cm wide, and 4 cm high. What is its surface area?

- A) 47 cm^2
- B) 120 cm^2
- C) 60 cm^2
- D) 94 cm^2

185. A circle has a circumference of 31.4 cm. Using 3.14 for π , what is its diameter?

- A) 5 cm
- B) 10 cm
- C) 6.28 cm
- D) 15.7 cm

186. A hiking trail is 2.4 kilometers long. How many meters long is it?

- A) 24
- B) 2,400
- C) 24,000
- D) 240

187. A scale drawing uses 1 inch to represent 6 feet. A wall measures 4.5 inches on the drawing. How long is the actual wall?

- A) 27 feet
- B) 24 feet
- C) 30 feet
- D) 36 feet

188. The numbers 12, 15, 18, 18, and 22 are recorded. What is the median?

- A) 15
- B) 18
- C) 17
- D) 22

189. A spinner has 8 equal sections: 3 yellow, 2 purple, and 3 orange. What is the probability of landing on a section that is not purple?

- A) $\frac{3}{4}$
- B) $\frac{3}{8}$
- C) $\frac{1}{4}$
- D) $\frac{5}{8}$

190. The mean of four scores is 18. Three of the scores are 12, 19, and 21. What is the fourth score?

- A) 18
- B) 20
- C) 22
- D) 19

191. What is the least common multiple of 8 and 12?

- A) 16
- B) 48
- C) 20
- D) 24

192. Which value is greatest?

- A) 0.82
- B) $\frac{5}{6}$
- C) 81%
- D) $\frac{4}{5}$

193. What is $2\frac{3}{4} + 1\frac{5}{8}$?

- A) $4\frac{1}{8}$
- B) $3\frac{7}{8}$
- C) $4\frac{3}{8}$
- D) $4\frac{5}{8}$

194. If $f(n) = 3n - 2$, what is $f(7)$?

- A) 17
- B) 19
- C) 23
- D) 21

195. Solve: $2(x - 3) + 5 = 17$

- A) 9
- B) 6
- C) 7
- D) 8

196. Which inequality is equivalent to $5x + 4 < 24$?

- A) $x > 4$
- B) $x < 20$
- C) $x < 4$
- D) $x > 20$

197. A machine fills 18 bottles in 3 minutes at a constant rate. At that rate, how many bottles will it fill in 8 minutes?

- A) 56
- B) 42
- C) 54
- D) 48

198. A right triangle has legs of 6 cm and 8 cm. What is the length of its hypotenuse?

- A) 9 cm
- B) 10 cm
- C) 14 cm
- D) 12 cm

199. A parallelogram has a base of 11 cm and a height of 7 cm. What is its area?

- A) 77 cm^2
- B) 36 cm^2
- C) 63 cm^2
- D) 154 cm^2

200. Two similar triangles have corresponding side lengths 4 cm and 10 cm. If another side of the smaller triangle is 6 cm, what is the corresponding side of the larger triangle?

- A) 15 cm
- B) 16 cm
- C) 12 cm
- D) 18 cm

201. A container holds 3.5 liters. How many milliliters is that?

- A) 35,000
- B) 35
- C) 350
- D) 3,500

202. A rectangular tank measures 40 cm by 25 cm by 30 cm. What is its volume in liters?

- A) 30 liters
- B) 3 liters
- C) 300 liters
- D) 3,000 liters

203. A 12-ounce bag costs \$3.60. What is the cost per ounce?

- A) \$0.20
- B) \$0.30
- C) \$0.25
- D) \$0.36

204. The data set is 6, 9, 11, 14, 14, 18. What is the range?

- A) 8
- B) 9
- C) 12
- D) 24

205. A bag contains 4 red, 5 blue, and 3 green tiles. One tile is chosen at random. What is the probability that it is green or red?

- A) $\frac{2}{3}$
- B) $\frac{5}{12}$
- C) $\frac{7}{12}$
- D) $\frac{3}{4}$

206. A student's four quiz scores are 78, 84, 91, and 87. What score on a fifth quiz would make the mean exactly 86?

- A) 86
- B) 88
- C) 92
- D) 90

207. What is $|-23| + |-8|$?

- A) 15
- B) 31
- C) -31
- D) 184

208. Evaluate: $18 - 2 \times (5 + 1)$

- A) 16
- B) 12
- C) 6
- D) 96

209. Which expression is equivalent to $\frac{7}{12} + \frac{5}{18}$?

- A) $\frac{11}{30}$
- B) $\frac{23}{36}$
- C) $1 \frac{1}{36}$
- D) $\frac{31}{36}$

210. A store marks an \$84 item down by 30%. What is the sale price?

- A) \$25.20
- B) \$54.00
- C) \$58.80
- D) \$64.80

211. Simplify: $4a + 3 - 2a + 7$

- A) $2a + 4$
- B) $2a + 10$
- C) $6a + 10$
- D) $6a + 4$

212. A taxi fare is modeled by $C = 4 + 2.5m$, where m is the number of miles. What does 4 represent?

- A) The number of miles traveled
- B) The cost per mile
- C) The initial charge before mileage is added
- D) The total fare after four miles

213. Three consecutive integers have a sum of 72. What is the largest integer?

- A) 23
- B) 25
- C) 26
- D) 24

214. Which expression represents “five less than twice a number n ”?

- A) $5 - 2n$
- B) $2(n - 5)$
- C) $2n - 5$
- D) $5n - 2$

215. A circle has diameter 14 cm. Using $\pi = 22/7$, what is its circumference?

- A) 22 cm
- B) 88 cm
- C) 154 cm
- D) 44 cm

216. The interior angles of a quadrilateral measure 90° , 85° , 110° , and x° . What is x ?

- A) 65°
- B) 75°
- C) 70°
- D) 85°

217. A rectangular box is 6 cm long, 4 cm wide, and 3 cm high. What is its total surface area?

- A) 54 cm^2
- B) 72 cm^2
- C) 108 cm^2
- D) 96 cm^2

218. On a map, 1 inch represents 18 miles. Two towns are 3.5 inches apart on the map. What is the actual distance?

- A) 54 miles
- B) 60 miles
- C) 63 miles
- D) 72 miles

219. A ribbon 9 feet long is cut into pieces that are each 18 inches long. How many complete pieces can be cut?

- A) 6
- B) 5
- C) 4
- D) 9

220. The numbers of pages read by five students are 18, 24, 24, 31, and 38. What is the median?

- A) 38
- B) 27
- C) 31
- D) 24

221. A spinner has 8 equal sections numbered 1 through 8. What is the probability of spinning an even number and then, on a second spin, a number greater than 6?

- A) $\frac{1}{16}$
- B) $\frac{1}{8}$
- C) $\frac{3}{8}$
- D) $\frac{1}{4}$

222. Set A is 4, 4, 5, 7, 10. Set B is 1, 5, 6, 6, 12. Which statement is true?

- A) The sets have the same mean and the same range.
- B) Set B has a greater mean, but Set A has a greater range.
- C) Set A has a greater mean and a greater range.
- D) The sets have the same mean, but Set B has a greater range.

223. Which value is greatest?

- A) $\frac{5}{8}$
- B) 62%
- C) 0.61
- D) 0.6

224. What is the least common multiple of 14 and 21?

- A) 28
- B) 35
- C) 42
- D) 84

225. A number rounds to 4.7 when rounded to the nearest tenth. Which could be the original number?

- A) 4.66
- B) 4.75
- C) 4.81
- D) 4.64

226. For $f(x) = 3x - 4$, what is $f(-2)$?

- A) 2
- B) -6
- C) -10
- D) 10

227. A number x satisfies $3(x + 2) = 2x + 11$. What is x ?

- A) 3
- B) 7
- C) 5
- D) 9

228. Which inequality represents all numbers x that are at least 4 but less than 9?

- A) $4 < x \leq 9$
- B) $4 < x < 9$
- C) $x \leq 4$ or $x > 9$
- D) $4 \leq x < 9$

229. A tank contains 18 gallons and is filled at 4 gallons per minute. Which expression gives the amount after t minutes?

- A) $18t + 4$
- B) $4t - 18$
- C) $18 + 4t$
- D) $22t$

230. A right triangle has legs 9 cm and 12 cm. What is its hypotenuse?

- A) 15 cm
- B) 13 cm
- C) 18 cm
- D) 21 cm

231. A regular hexagon has a perimeter of 54 cm. What is one side length?

- A) 9 cm
- B) 8 cm
- C) 6 cm
- D) 12 cm

232. Two similar rectangles have corresponding sides 6 cm and 15 cm. If another side of the smaller rectangle is 8 cm, what is the corresponding larger side?

- A) 12 cm
- B) 24 cm
- C) 17 cm
- D) 20 cm

233. A movie begins at 6:47 p.m. and ends at 9:12 p.m. How long does it last?

- A) 2 hours 15 minutes
- B) 2 hours 35 minutes
- C) 3 hours 5 minutes
- D) 2 hours 25 minutes

234. A container holds 2.5 quarts. How many cups does it hold? Use 1 quart = 4 cups.

- A) 10 cups
- B) 6 cups
- C) 8 cups
- D) 12 cups

235. A cyclist travels 42 miles in 3.5 hours at a constant rate. What is the average speed?

- A) 10 mph
- B) 14 mph
- C) 15 mph
- D) 12 mph

236. A bag contains 5 red, 3 blue, and 2 yellow tokens. What is the probability a randomly selected token is not blue?

- A) $\frac{3}{10}$
- B) $\frac{7}{10}$
- C) $\frac{1}{2}$
- D) $\frac{4}{5}$

237. The mean of four test scores is 82. Three scores are 76, 81, and 88. What is the fourth score?

- A) 79
- B) 83
- C) 85
- D) 81

238. A survey records 12 votes for A, 18 for B, 10 for C, and 20 for D. What fraction were for B?

- A) $\frac{3}{8}$
- B) $\frac{1}{3}$
- C) $\frac{3}{10}$
- D) $\frac{2}{5}$

Section 5: Language

60 questions | Recommended time: 30 minutes

239. Which sentence is punctuated correctly?

- A) The box contained three items a compass: a flashlight, and a whistle.
- B) The box contained: three items, a compass, a flashlight, and a whistle.
- C) The box contained three items; a compass, a flashlight, and a whistle.
- D) The box contained three items: a compass, a flashlight, and a whistle.

240. Which sentence is punctuated correctly?

- A) "I found the missing folder", Elena said.
- B) "I found the missing folder," Elena said.
- C) "I found the missing folder" Elena, said.
- D) "I found, the missing folder," Elena said.

241. Which sentence correctly uses a semicolon?

- A) The rain stopped; and the players returned to the field.
- B) The rain; stopped, and the players returned to the field.
- C) The rain stopped the players; returned to the field.
- D) The rain stopped; the players returned to the field.

242. Which sentence uses capitalization correctly?

- A) Our class begins Spanish lessons in September.
- B) Our Class begins spanish lessons in September.
- C) Our class begins Spanish lessons in september.
- D) Our class begins spanish Lessons in September.

243. Which sentence uses capitalization correctly?

- A) The club will meet on memorial Day.
- B) The club will meet on Memorial Day.
- C) The Club will meet on Memorial day.
- D) The club will meet on memorial day.

244. Which word is spelled correctly?

- A) embarass
- B) embarrasss
- C) embaras
- D) embarrass

245. Which sentence contains a misspelled word?

- A) The mechanic inspected the equipment carefully.
- B) The audience remained silent during the announcement.
- C) The scientist recorded the occurence in her notes.
- D) The package arrived at a convenient time.

246. Which sentence has correct subject-verb agreement?

- A) A basket of fresh peaches sit on the counter.
- B) Each of the volunteers brings a reusable bottle.
- C) The list of supplies are on the desk.
- D) The collection of stamps belong to my uncle.

247. Which sentence uses the correct verb form?

- A) By the time we arrived, the movie had begun.
- B) By the time we arrived, the movie has began.
- C) By the time we arrived, the movie had began.
- D) By the time we arrived, the movie begun.

248. Which sentence places the modifier correctly?

- A) Walking through the station, the announcement surprised Lena.
- B) Lena heard an announcement walking through the station.
- C) Walking through the station, Lena heard an announcement.
- D) The announcement, walking through the station, surprised Lena.

249. Which sentence uses the correct word?

- A) The new route is shorter than the old one.
- B) The new route is more short than the old one.
- C) The new route is shortest then the old one.
- D) The new route is shorter then the old one.

250. Which sentence has the clearest pronoun reference?

- A) When the dog followed the runner, it was tired.
- B) When Leo spoke with Marcus, he said the schedule had changed.
- C) After the coach called the captain, he checked the roster.
- D) Maya told Priya that Priya's notebook was on the table.

Editing Passage: A Useful School Swap Shelf

(1) Students at Ridgeway School often finished notebooks with several blank pages remaining. (2) Others had binders, folders, or art supplies they no longer needed. (3) The student council proposed a swap shelf where usable school materials could be left for other students. (4) At first, some students placed broken pens and heavily damaged folders on the shelf. (5) The council responded by posting a short list of acceptable items. (6) It also added labeled containers so supplies would be easier to sort. (7) Within a month, students were regularly taking and leaving useful materials. (8) The shelf did not eliminate the need to buy school supplies, but it helped usable items remain in circulation longer.

251. Which sentence would be the best topic sentence to place before sentence 1?

- A) School supplies are available in many colors.
- B) A simple exchange system helped students reuse school materials.
- C) Ridgeway School has several long hallways.
- D) Some students prefer notebooks to binders.

252. Which sentence could best be added after sentence 6 to improve the transition to the result in sentence 7?

- A) The shelf was made of wood and stood near a window.
- B) Many classrooms also had desks and cabinets.
- C) These changes made the shelf easier for students to use responsibly.
- D) The council met on a Tuesday afternoon.

253. Which revision best combines sentences 1 and 2 while preserving their meaning?

- A) Students had blank notebook pages, but binders are supplies.
- B) Some students had notebooks with blank pages remaining, while others had binders, folders, or art supplies they no longer needed.
- C) Students had school supplies, which were sometimes different.
- D) Students finished notebooks, and others no longer needed.

254. Which sentence uses the apostrophe correctly?

- A) The two cyclists helmets were stored inside.
- B) The two cyclist's helmets were stored inside.
- C) The two cyclists' helmets were stored inside.
- D) The two cyclists's helmets were stored inside.

255. Which sentence is punctuated correctly?

- A) After, the final bell rang we hurried to the bus.
- B) After the final bell rang we hurried to the bus.
- C) After the final bell rang, we hurried to the bus.
- D) After the final bell, rang we hurried to the bus.

256. Which sentence uses capitalization correctly?

- A) My Aunt moved from ohio to the Pacific northwest.
- B) My aunt moved from Ohio to the Pacific Northwest.
- C) My aunt moved from ohio to the Pacific Northwest.
- D) My aunt moved from Ohio to the pacific Northwest.

257. Which sentence uses capitalization correctly?

- A) We read a poem by Professor Elena Ruiz in English class.
- B) We read a poem by professor Elena Ruiz in English Class.
- C) We read a poem by Professor elena Ruiz in english class.
- D) We read a poem by professor Elena ruiz in English class.

258. Which word is spelled correctly?

- A) entreprenuer
- B) entrepreneur
- C) entreperneur
- D) entrepreneuer

259. Which sentence contains a misspelled word?

- A) The speaker gave a persistant reminder.
- B) The committee reached a separate decision.
- C) The athlete followed a regular schedule.
- D) The librarian recommended a reference book.

260. Which sentence contains no spelling errors?

- A) The principal made an immediate announcement.
- B) The gardener used fertalizer carefully.
- C) The musician practiced the rythm slowly.
- D) The restaurant had a noticable entrance.

261. Which sentence uses pronoun case correctly?

- A) The coach gave the forms to Jordan and I.
- B) The coach gave the forms to Jordan and me.
- C) The coach gave I and Jordan the forms.
- D) The coach gave me and Jordan's the forms.

262. Which sentence is a complete sentence?

- A) Because the trail was muddy after the storm.
- B) The volunteers who arrived before sunrise.
- C) While the workers repaired the bridge.
- D) The volunteers moved the supplies into the shelter.

263. Which sentence uses parallel structure correctly?

- A) The job requires to sort files, answering calls, and greeting visitors.
- B) The job requires sorting files, answering calls, and to greet visitors.
- C) The job requires sorting files, answering calls, and greeting visitors.
- D) The job requires sorting files, to answer calls, and visitors are greeted.

264. Which sentence uses the correct word?

- A) Please accept this gift as a token of thanks.
- B) Please acceptance this gift as a token of thanks.
- C) Please accepting this gift as a token of thanks.
- D) Please except this gift as a token of thanks.

265. Which sentence is most concise?

- A) The meeting was postponed until a later time in the future.
- B) The meeting was postponed until later in the future ahead.
- C) The meeting was postponed until later.
- D) The meeting was postponed and delayed until a later time.

Editing Passage: A Hallway History Display

(1) The eighth-grade history club wanted more students to notice its collection of neighborhood photographs. (2) The photographs had previously been kept in a cabinet and used mainly during club meetings. (3) Members decided to create a temporary hallway display. (4) They grouped the photographs by decade and wrote brief captions explaining visible changes in streets, stores, and public spaces. (5) One member suggested adding a long paragraph about the history of photography equipment. (6) The group decided that this information would distract from the display's focus on neighborhood change. (7) After the display opened, several students stopped

to compare familiar locations with their earlier appearance. (8) The club then added a notebook where viewers could record questions or identify places they recognized.

266. Which revision of sentence 2 is clearest and most concise?

- A) Previously, the photographs were kept in a cabinet and used mainly during club meetings.
- B) The cabinet was where photographs that the club mainly used were previously being kept.
- C) The photographs, previously before, had been kept in a cabinet for club meetings mainly.
- D) Mainly club meetings used photographs previously kept inside of a cabinet.

267. Which sentence could best be added after sentence 4?

- A) The school cafeteria served lunch nearby.
- B) These captions helped viewers connect each image with a specific period of local change.
- C) Some club members also enjoyed science class.
- D) Photography equipment can be expensive.

268. Why is sentence 5 useful to the paragraph?

- A) It proves that photography equipment is the display's main subject.
- B) It explains why the photographs were damaged.
- C) It provides the paragraph's final conclusion.
- D) It introduces an idea the group evaluates and rejects to preserve the display's focus.

269. Which sentence is punctuated correctly?

- A) The supply box contained three items; tape, scissors, and string.
- B) The supply box contained three items: tape, scissors, and string.
- C) The supply box: contained three items, tape, scissors, and string.
- D) The supply box contained: three items tape, scissors, and string.

270. Which sentence is punctuated correctly?

- A) Nora finished the poster, however, she forgot to add the date.
- B) Nora finished the poster; however, she forgot to add the date.
- C) Nora finished the poster however; she forgot to add the date.
- D) Nora finished the poster however she forgot; to add the date.

271. Which sentence uses quotation marks and punctuation correctly?

- A) "Please close the window", said Mr. Lane.
- B) "Please close the window" said, Mr. Lane.
- C) "Please close the window," said Mr. Lane.
- D) Please "close the window," said Mr. Lane.

272. Which sentence uses capitalization correctly?

- A) Our Class visits the national air and space museum in October.
- B) Our class visits the National Air And Space Museum in october.
- C) Our class visits the National Air and Space Museum in October.
- D) Our class visits the national Air and Space museum in October.

273. Which sentence uses capitalization correctly?

- A) On memorial Day, the scouts met near Lake Crescent.
- B) On memorial day, the Scouts met near Lake crescent.
- C) On Memorial day, the scouts met near lake Crescent.
- D) On Memorial Day, the scouts met near Lake Crescent.

274. Which word is spelled correctly?

- A) accomodate
- B) accomodate
- C) accomadate
- D) accommodate

275. Which sentence contains no spelling errors?

- A) The audience showed genuin enthusiasm.
- B) The audiance showed genuine enthusiasm.
- C) The audience showed genuine enthusiasm.
- D) The audience showed genuine enthusiasim.

276. Which sentence uses subject-verb agreement correctly?

- A) Neither of the two routes leads to the station.
- B) Neither of the two routes are leading to the station.
- C) Neither of the two routes lead to the station.
- D) Neither of the two routes have led to the station every day.

277. Which sentence maintains consistent verb tense?

- A) Yesterday, Mia had packed her bag and walks to the bus stop.
- B) Yesterday, Mia packed her bag and walks to the bus stop.
- C) Yesterday, Mia packs her bag and walked to the bus stop.
- D) Yesterday, Mia packed her bag and walked to the bus stop.

278. Which sentence places the modifier correctly?

- A) Maya photographed the branches covered in frost.
- B) Covered in frost, the camera photographed the branches for Maya.
- C) The branches photographed Maya while covered in frost.
- D) Covered in frost, Maya photographed the branches.

279. Which sentence uses the correct word?

- A) The new rule may effect how teams schedule practice.
- B) The new rule may affection how teams schedule practice.
- C) The new rule may affect how teams schedule practice.
- D) The new rule may effected how teams schedule practice.

280. Which sentence uses the most precise word?

- A) The scientist looked at the sample.
- B) The scientist examined the sample.
- C) The scientist did something with the sample.
- D) The scientist was around the sample.

281. Which transition best completes the sentence? The first route is shorter. _____, the second route avoids the steep hill.

- A) Therefore
- B) For example
- C) Similarly
- D) However

Editing Passage: A School Seed Exchange

(1) The environmental club organized a seed exchange so students could share extra garden seeds instead of discarding them. (2) Members sorted packets into vegetables, herbs, and flowers and placed simple growing notes beside each group. (3) At first, the club planned to accept every packet that students brought. (4) They later decided to exclude packets with missing labels because no one could reliably identify their contents. (5) The exchange opened during lunch on Thursday. (6) By the end of the period, students had traded dozens of packets and left suggestions for plants they hoped to see at the next exchange. (7) The club used those suggestions when planning a second event for spring.

282. Which sentence would best follow sentence 2?

- A) Some students prefer sandwiches for lunch.
- B) Gardening has existed for thousands of years.
- C) The labels helped students choose seeds suited to the plants they wanted to grow.
- D) The club's meeting room had four windows.

283. Which is the best way to combine sentences 6 and 7?

- A) Students traded packets, and suggestions were there, and spring was later planned by the club.
- B) The club used suggestions because students had packets and the period ended.
- C) By the end of the period, students traded dozens of packets; spring is a season.
- D) By the end of the period, students had traded dozens of packets and left suggestions, which the club used to plan a second exchange for spring.

284. Which sentence is punctuated correctly?

- A) After the final bell rang we hurried to the bus.
- B) After, the final bell rang we hurried to the bus.
- C) After the final bell rang, we hurried to the bus.
- D) After the final bell, rang we hurried to the bus.

285. Which sentence uses the apostrophe correctly?

- A) The two coach's offices are beside the gym.
- B) The two coaches' offices are beside the gym.
- C) The two coaches's offices are beside the gym.
- D) The two coaches offices' are beside the gym.

286. Which sentence uses capitalization correctly?

- A) My aunt studies Spanish and history at Western State University.
- B) My aunt studies spanish and History at Western State University.
- C) My Aunt studies Spanish and history at western state university.
- D) My aunt studies Spanish and History at Western state University.

287. Which sentence uses capitalization correctly?

- A) Professor Elena Ruiz will speak at the City Arts Council meeting.
- B) Professor Elena ruiz will speak at the City arts council meeting.
- C) professor Elena Ruiz will speak at the city Arts Council meeting.
- D) Professor Elena Ruiz will speak at the city arts Council meeting.

288. Which word is spelled correctly?

- A) priviledge
- B) privilege
- C) privelege
- D) privilege

289. Which word is spelled correctly?

- A) maintenance
- B) maintenence
- C) maintnance
- D) maintainance

290. Which sentence contains no spelling errors?

- A) The committee reached a unanimous decision.
- B) The commitee reached a unanimous decision.
- C) The committee reached a unaminous decision.
- D) The committee reached a unanimous descision.

291. Which sentence uses pronoun case correctly?

- A) The coach gave Maya and I the schedule.
- B) The coach gave I and Maya the schedule.
- C) The coach gave me and she the schedule.
- D) The coach gave Maya and me the schedule.

292. Which sentence is complete rather than a fragment?

- A) Because the storm ended before noon.
- B) Although the equipment was already packed.
- C) While waiting beside the locked gate.
- D) The volunteers returned to the field after the storm ended.

293. Which sentence uses parallel structure correctly?

- A) The job requires sorting files, answering calls, and to greet visitors.
- B) The job requires to sort files, answering calls, and greeting visitors.
- C) The job requires sorting files, answering calls, and greeting visitors.
- D) The job requires sorting files, to answer calls, and visitors are greeted.

294. Which sentence is most concise?

- A) At this point in time, the team is currently ready to begin.
- B) The team is ready to begin.
- C) The team is currently ready to begin now.
- D) The team is ready at the present time to begin now.

295. Which sentence has the clearest pronoun reference?

- A) Lena, who was worried, spoke to Ava.
- B) When Lena spoke to Ava, this made her worried.
- C) When Lena spoke to Ava, she was worried.
- D) She spoke to Ava because Lena was worried.

296. Which transition best completes the sentence? The first experiment produced inconsistent results. _____, the team revised its procedure before trying again.

- A) Meanwhile
- B) Likewise
- C) For instance
- D) As a result

Editing Passage: The Hallway Sound Survey

(1) A student council committee wanted to learn which school hallways became noisiest between classes. (2) Instead of relying on memory, members stood at six locations and recorded sound levels during three passing periods. (3) They also noted nearby features, such as stairways, lockers, and water fountains. (4) One hallway produced high readings only during the final passing period, when two after-school groups gathered near its entrance. (5) The committee concluded that location alone did not explain every result. (6) In its report, the committee recommended repeating the survey on several different days before suggesting any changes to hallway routines. (7) This additional data would help show whether the same patterns appeared consistently.

297. Which sentence would best serve as a topic sentence before sentence 1?

- A) The school building has several classrooms with windows.
- B) Many students carry notebooks between classes.
- C) After-school clubs meet for many different reasons.
- D) Students often have strong opinions about noise, but the committee chose to investigate the issue with recorded evidence.

298. Which revision best combines sentences 6 and 7?

- A) The report suggested changes because different days have patterns.
- B) The committee recommended several days, and patterns were consistent, and changes were suggested.
- C) The committee recommended repeating the survey on several different days so it could determine whether the same patterns appeared consistently before suggesting changes.
- D) Repeating the survey was recommended; hallways have routines.

Answer Key

Question numbers continue across all five sections.

Verbal Skills

1. D	2. B	3. A	4. B	5. C	6. D	7. C	8. D
9. D	10. B	11. A	12. C	13. A	14. D	15. B	16. B
17. A	18. B	19. B	20. D	21. A	22. A	23. C	24. B
25. D	26. D	27. C	28. C	29. A	30. B	31. A	32. A
33. C	34. D	35. C	36. B	37. C	38. B	39. A	40. A
41. B	42. C	43. D	44. A	45. C	46. D	47. D	48. A
49. C	50. C	51. D	52. D	53. C	54. B	55. B	56. D
57. A	58. A	59. B	60. C				

Quantitative Skills

61. B	62. C	63. B	64. B	65. C	66. A	67. A	68. D
69. C	70. A	71. D	72. A	73. B	74. C	75. A	76. C
77. D	78. B	79. D	80. C	81. C	82. B	83. D	84. A
85. C	86. A	87. D	88. C	89. D	90. C	91. B	92. D
93. C	94. B	95. D	96. C	97. B	98. A	99. B	100. A
101. A	102. B	103. B	104. D	105. A	106. C	107. D	108. A
109. A	110. D	111. D	112. B				

Reading Comprehension

113. B	114. A	115. A	116. B	117. B	118. D	119. C	120. A
121. D	122. C	123. B	124. B	125. C	126. D	127. B	128. C
129. D	130. C	131. B	132. C	133. D	134. B	135. D	136. D
137. C	138. B	139. B	140. A	141. D	142. A	143. C	144. C
145. A	146. D	147. B	148. D	149. A	150. A	151. D	152. D
153. A	154. B	155. D	156. C	157. C	158. D	159. C	160. C
161. A	162. A	163. B	164. A	165. B	166. A	167. B	168. A
169. B	170. A	171. C	172. C	173. D	174. A		

Mathematics

175. A	176. B	177. D	178. A	179. D	180. C	181. A	182. A
183. D	184. D	185. B	186. B	187. A	188. B	189. A	190. B
191. D	192. B	193. C	194. B	195. A	196. C	197. D	198. B
199. A	200. A	201. D	202. A	203. B	204. C	205. C	206. D

207. B	208. C	209. D	210. C	211. B	212. C	213. B	214. C
215. D	216. B	217. C	218. C	219. A	220. D	221. B	222. D
223. A	224. C	225. A	226. C	227. C	228. D	229. C	230. A
231. A	232. D	233. D	234. A	235. D	236. B	237. B	238. C

Language

239. D	240. B	241. D	242. A	243. B	244. D	245. C	246. B
247. A	248. C	249. A	250. D	251. B	252. C	253. B	254. C
255. C	256. B	257. A	258. B	259. A	260. A	261. B	262. D
263. C	264. A	265. C	266. A	267. B	268. D	269. B	270. B
271. C	272. C	273. D	274. D	275. C	276. A	277. D	278. A
279. C	280. B	281. D	282. C	283. D	284. C	285. B	286. A
287. A	288. B	289. A	290. A	291. D	292. D	293. C	294. B
295. A	296. D	297. D	298. C				

Detailed Explanations

Verbal Skills

1. HSPT-V7-001 - Answer D: adaptable

Resilient describes an ability to recover or adjust successfully, making adaptable the closest choice.

2. HSPT-V7-002 - Answer B: careful

Meticulous means very careful and attentive to details.

3. HSPT-V7-003 - Answer A: plentiful

Abundant means existing in large quantities, so plentiful is the closest synonym.

4. HSPT-V7-004 - Answer B: flexible

Rigid means stiff or inflexible, making flexible its clearest opposite.

5. HSPT-V7-005 - Answer C: abundant

Scarce means limited or hard to find; abundant means plentiful.

6. HSPT-V7-006 - Answer D: reveal

Conceal means to hide, while reveal means to make known or visible.

7. HSPT-V7-007 - Answer C: horse

A sapling is a young tree, just as a colt is a young horse.

8. HSPT-V7-008 - Answer D: fabric

Clay can be made into pottery, while wool can be made into fabric.

9. HSPT-V7-009 - Answer D: temperature

A compass indicates direction, while a thermometer indicates temperature.

10. HSPT-V7-010 - Answer B: novel

Sonnet, haiku, and limerick are poetic forms; a novel is extended prose fiction.

11. HSPT-V7-011 - Answer A: silver

Copper, iron, aluminum, and silver are metals.

12. HSPT-V7-012 - Answer C: aquarium

Orchard, vineyard, and plantation are places where plants or crops are cultivated; an aquarium is not.

13. HSPT-V7-013 - Answer A: J, K, L, M

This schedule places K after J and before M, and L is not first.

14. HSPT-V7-014 - Answer D: Tree T is evergreen.

Tree T is a cedar in the grove, and every cedar in that grove is stated to be evergreen.

15. HSPT-V7-015 - Answer B: blue, amber, green

Blue must be above amber, and green must be below amber, fixing this top-to-bottom order.

16. HSPT-V7-016 - Answer B: cautious

Prudent means showing careful judgment, so cautious is the closest synonym.

17. HSPT-V7-017 - Answer A: reduce

Diminish means to make or become smaller, so reduce is the closest choice.

18. HSPT-V7-018 - Answer B: unbiased

Impartial means not favoring one side, which is equivalent to unbiased.

19. HSPT-V7-019 - Answer B: simple

Intricate means complex or highly detailed; simple is its clearest opposite.

20. HSPT-V7-020 - Answer D: eager

Reluctant means unwilling or hesitant, while eager describes readiness or strong willingness.

21. HSPT-V7-021 - Answer A: contract

To expand is to become larger; to contract is to become smaller.

22. HSPT-V7-022 - Answer A: account

A key provides access through a lock, while a password provides access to an account.

23. HSPT-V7-023 - Answer C: burrow

A hive is a bee's dwelling, while a burrow is a rabbit's dwelling.

24. HSPT-V7-024 - Answer B: vehicle
An editor works on a manuscript, just as a mechanic works on a vehicle.
25. HSPT-V7-025 - Answer D: flute
Violin, cello, and viola are string instruments; a flute is a wind instrument.
26. HSPT-V7-026 - Answer D: hyacinth
Tulip, daffodil, crocus, and hyacinth are flowering plants that grow from bulbs or corm-like underground structures.
27. HSPT-V7-027 - Answer C: copper
Ruby, emerald, and sapphire are gemstones; copper is a metal.
28. HSPT-V7-028 - Answer C: The museum did not open late.
By contraposition, if a late opening would cancel the first tour and it was not canceled, the museum did not open late.
29. HSPT-V7-029 - Answer A: R
Q must be below R, and P cannot be on top, so R is the only box that can occupy the top position.
30. HSPT-V7-030 - Answer B: None of the first three conclusions must be true.
The two statements do not establish whether any eighth grader attending the meeting belongs to the robotics team.
31. HSPT-V7-031 - Answer A: limited
Limited is closest because scarce means available only in a small amount.
32. HSPT-V7-032 - Answer A: confirm
To verify is to establish that something is true or accurate, so confirm is the closest synonym.
33. HSPT-V7-033 - Answer C: able to recover
Resilient describes something able to recover or return to form after difficulty or pressure.
34. HSPT-V7-034 - Answer D: flexible
Rigid means stiff or inflexible, so flexible is the clearest opposite.
35. HSPT-V7-035 - Answer C: reveal
To conceal is to hide; to reveal is to make known or visible.
36. HSPT-V7-036 - Answer B: scarce
Abundant means plentiful, while scarce means present in insufficient or limited quantity.
37. HSPT-V7-037 - Answer C: food
An archive stores records, while a pantry stores food.
38. HSPT-V7-038 - Answer B: brick
Flour is a material used to make bread, just as clay can be a material used to make a brick.
39. HSPT-V7-039 - Answer A: approval
An alarm expresses or provides a warning; applause expresses approval.
40. HSPT-V7-040 - Answer A: daisy
Cedar, maple, and oak are trees; a daisy is a flowering herbaceous plant.
41. HSPT-V7-041 - Answer B: limestone
Quartz, granite, marble, and limestone are naturally occurring mineral or rock materials; the others are manufactured or organic materials.
42. HSPT-V7-042 - Answer C: sprint
Stroll, saunter, and amble describe walking leisurely, while sprint describes running quickly.
43. HSPT-V7-043 - Answer D: L
Because L is before J, J is immediately before K, and M is after K, the only possible order begins with L.
44. HSPT-V7-044 - Answer A: weaving
After Monday leaves Tuesday or Wednesday, and excluding drawing leaves only Wednesday's weaving workshop.
45. HSPT-V7-045 - Answer C: No cedar box is on the bottom shelf.
Every cedar box is labeled, and no labeled box is on the bottom shelf, so no cedar box can be there.
46. HSPT-V7-046 - Answer D: honest
Candid means truthful and straightforward, so honest is the closest synonym.
47. HSPT-V7-047 - Answer D: possible
Feasible means capable of being done successfully, so possible is the closest synonym.
48. HSPT-V7-048 - Answer A: watchful
Vigilant means staying alert and watchful for possible problems or changes.

49. HSPT-V7-049 - Answer C: restless

Tranquil means calm and peaceful, while restless describes a lack of calm.

50. HSPT-V7-050 - Answer C: temporary

Permanent means intended to last, while temporary means lasting only for a limited time.

51. HSPT-V7-051 - Answer D: vague

Precise means exact and clearly defined; vague means unclear or imprecise.

52. HSPT-V7-052 - Answer D: score

An architect creates a blueprint as a plan for construction; a composer creates a score as a plan for musical performance.

53. HSPT-V7-053 - Answer C: play

A chapter is a component of a book, just as a scene is a component of a play.

54. HSPT-V7-054 - Answer B: plant

A spark can initiate a fire, just as a seed can develop into a plant.

55. HSPT-V7-055 - Answer B: comet

Mercury, Venus, and Mars are planets; a comet is a different type of solar-system object.

56. HSPT-V7-056 - Answer D: kite

A rectangle, rhombus, trapezoid, and kite are all quadrilaterals.

57. HSPT-V7-057 - Answer A: expert

Novice, beginner, and apprentice can describe someone still learning; expert describes someone highly skilled.

58. HSPT-V7-058 - Answer A: vase

The required order is model ship, globe, clock, vase, so the vase must occupy the bottom shelf.

59. HSPT-V7-059 - Answer B: Priya attends Thursday.

Club members attend exactly one session; since Priya does not attend Tuesday, she must attend Thursday.

60. HSPT-V7-060 - Answer C: green

The only arrangement satisfying the conditions places green first and red second.

Quantitative Skills

61. HSPT-Q7-001 - Answer B: 57

Differences are +4, +7, +10, +13, so the next difference is +16 and $41 + 16 = 57$.

62. HSPT-Q7-002 - Answer C: 15

The operations alternate $\div 2$ and $+4$; $30 \div 2 = 15$.

63. HSPT-Q7-003 - Answer B: 20

Two interwoven sequences are 2, 4, 6, 8 and 5, 10, 15, 20.

64. HSPT-Q7-004 - Answer B: $1\frac{1}{2}$

Each term increases by $\frac{1}{4}$, so $1\frac{1}{4} + \frac{1}{4} = 1\frac{1}{2}$.

65. HSPT-Q7-005 - Answer C: 60

If 125% of the original equals 75, the original is $75 \div 1.25 = 60$.

66. HSPT-Q7-006 - Answer A: $\frac{3}{8}$

$3 \div 8 = 0.375$.

67. HSPT-Q7-007 - Answer A: 6

Multiplying the 5:2 ratio by 3 gives 15:6.

68. HSPT-Q7-008 - Answer D: Quantity B is greater.

Quantity A is 49, while Quantity B is 50.

69. HSPT-Q7-009 - Answer C: Quantity A is greater.

$y/3 = 9$, so $y = 27$, which is greater than 26.

70. HSPT-Q7-010 - Answer A: The two quantities are equal.

The greatest integer less than 12 is 11, equal to Quantity B.

71. HSPT-Q7-011 - Answer D: The two quantities are equal.

The rectangle area is 54 cm^2 and the triangle area is $\frac{1}{2} \times 12 \times 9 = 54\text{ cm}^2$.

72. HSPT-Q7-012 - Answer A: Quantity B is greater.

The pentagon perimeter is $5 \times 7 = 35\text{ cm}$, which is less than 36 cm.

73. HSPT-Q7-013 - Answer B: The two quantities are equal.

The cube volume is $4^3 = 64\text{ cm}^3$, and the prism volume is $8 \times 4 \times 2 = 64\text{ cm}^3$.

74. HSPT-Q7-014 - Answer C: 46

The amounts subtracted decrease by 1: -9, -8, -7, -6, so the next change is -5 and $51 - 5 = 46$.

75. HSPT-Q7-015 - Answer A: 2,160

The terms are multiplied successively by 2, 3, 4, 5, then 6; $360 \times 6 = 2,160$.

76. HSPT-Q7-016 - Answer C: -20

The decreases are 8, 15, 22, and 29; they grow by 7, so the next decrease is 36 and $16 - 36 = -20$.

77. HSPT-Q7-017 - Answer D: 54

A number divisible by both 6 and 9 must be divisible by 18; 54 is the only listed multiple of 18.

78. HSPT-Q7-018 - Answer B: 45

Multiply 27 by $\frac{5}{3}$: $27 \times \frac{5}{3} = 45$.

79. HSPT-Q7-019 - Answer D: 15%

The decrease is \$12, and $12 \div 80 = 0.15$, so the percent decrease is 15%.

80. HSPT-Q7-020 - Answer C: $\frac{3}{4}$ of 20

The values are 12, 12, 15, and 14 respectively, so $\frac{3}{4}$ of 20 is greatest.

81. HSPT-Q7-021 - Answer C: The two quantities are equal.

Both quantities equal 84.

82. HSPT-Q7-022 - Answer B: Quantity B is greater.

Solving $4x - 7 = 29$ gives $x = 9$, so Quantity B, 10, is greater.

83. HSPT-Q7-023 - Answer D: The two quantities are equal.

The least integer greater than -3 is -2, exactly equal to Quantity B.

84. HSPT-Q7-024 - Answer A: Quantity A is greater.

The square has side length 8 cm and area 64 cm^2 , which is greater than 60 cm^2 .

85. HSPT-Q7-025 - Answer C: Quantity B is greater.
The circle's diameter is 12 cm; the triangle's side is $39 \div 3 = 13$ cm, so Quantity B is greater.
86. HSPT-Q7-026 - Answer A: Quantity A is greater.
The rhombus area is one half of 10×12 , or 60 square centimeters; the triangle area is one half of 15×7 , or 52.5 square centimeters.
87. HSPT-Q7-027 - Answer D: 95
Each term is twice the preceding term plus 1; $47 \times 2 + 1 = 95$.
88. HSPT-Q7-028 - Answer C: 40
The subtractions are 4, 8, 12, and 16; subtracting the next multiple, 20, gives 40.
89. HSPT-Q7-029 - Answer D: 8
Two interwoven sequences are 1, 2, 4, 8 and 4, 8, 16, so the next odd-position term is 8.
90. HSPT-Q7-030 - Answer C: 2.6
The pattern alternates $\times 2$ and -0.2 ; after 2.8, subtract 0.2 to get 2.6.
91. HSPT-Q7-031 - Answer B: 8
The ratio is 2:3; multiplying both terms by 4 gives 8 cups of oats for 12 cups of nuts.
92. HSPT-Q7-032 - Answer D: $\frac{3}{8}$
0.375 equals $\frac{375}{1000}$, which simplifies to $\frac{3}{8}$.
93. HSPT-Q7-033 - Answer C: 72
If 125% of the original is 90, then the original is $90 \div 1.25 = 72$.
94. HSPT-Q7-034 - Answer B: The two quantities are equal.
Quantity A is $11 - 3 = 8$, and Quantity B is $2^3 = 8$.
95. HSPT-Q7-035 - Answer D: Quantity A is greater.
Solving $y/4 + 2 = 7$ gives $y = 20$, which is greater than 18.
96. HSPT-Q7-036 - Answer C: The relationship cannot be determined from the information given.
If m is below 7, Quantity B is greater; if $m = 7$ they are equal; if m exceeds 7, Quantity A is greater.
97. HSPT-Q7-037 - Answer B: Quantity B is greater.
The pentagon's perimeter is $5 \times 7 = 35$ cm, so Quantity B is greater.
98. HSPT-Q7-038 - Answer A: The two quantities are equal.
The prism's volume is $3 \times 4 \times 5 = 60$ cm³, equal to Quantity B.
99. HSPT-Q7-039 - Answer B: Quantity B is greater.
The exterior angle is $180^\circ - 112^\circ = 68^\circ$, so 70° is greater.
100. HSPT-Q7-040 - Answer A: 194
Each term is twice the preceding term minus 2; $98 \times 2 - 2 = 194$.
101. HSPT-Q7-041 - Answer A: 2
Each term is divided by 3, so $6 \div 3 = 2$.
102. HSPT-Q7-042 - Answer B: 16
The pattern alternates $+5$ and -1 ; after 17, subtract 1 to get 16.
103. HSPT-Q7-043 - Answer B: 45
If $0.4x = 18$, then $x = 18 \div 0.4 = 45$.
104. HSPT-Q7-044 - Answer D: 47
 $47 = 5 \times 9 + 2$, so it leaves a remainder of 2.
105. HSPT-Q7-045 - Answer A: 12
The quantity is 42, and $\frac{2}{7}$ of 42 is 12.
106. HSPT-Q7-046 - Answer C: 15%
The decrease is \$12, and $12 \div 80 = 0.15$, or 15%.
107. HSPT-Q7-047 - Answer D: The two quantities are equal.
Because $\frac{3}{5}$ equals 60%, both quantities are equal.
108. HSPT-Q7-048 - Answer A: Quantity A is greater.
Here n^2 is positive while n is negative, so Quantity A is greater.
109. HSPT-Q7-049 - Answer A: The relationship cannot be determined from the information given.
The condition fixes only the difference, so the sum is not determined.

110. HSPT-Q7-050 - Answer D: Quantity B is greater.

The triangle's area is 30 cm^2 , which is less than 32 cm^2 .

111. HSPT-Q7-051 - Answer D: The two quantities are equal.

Diameter is twice radius, so $2 \times 9 = 18 \text{ cm}$.

112. HSPT-Q7-052 - Answer B: The relationship cannot be determined from the information given.

The third side can vary between 4 and 16 cm, so its relation to 8 is not fixed.

Reading Comprehension

113. HSPT-R7-001 - Answer B: To keep track of sound-board settings

The passage says Nia marked each adjustment as she learned the sound board.

114. HSPT-R7-002 - Answer A: She sees that Nia has handled the problem effectively.

Nia lowers the faulty channel and switches microphones, allowing the rehearsal to continue.

115. HSPT-R7-003 - Answer A: an indication or reminder.

Nia comes to interpret nervousness as an indication to stay attentive rather than as proof of unpreparedness.

116. HSPT-R7-004 - Answer B: She comes to see nervousness as something that can sharpen her attention.

The final paragraph explicitly reframes nervousness as a signal to pay attention.

117. HSPT-R7-005 - Answer B: Confidence can grow from successfully responding to an unexpected problem.

Handling the microphone failure shows Nia that she can respond effectively under pressure.

118. HSPT-R7-006 - Answer D: To slow runoff and allow some water to soak into the ground

The passage identifies slowing and absorbing runoff as central functions of rain gardens.

119. HSPT-R7-007 - Answer C: quickly.

Rapidly means at a fast rate, so quickly fits the context.

120. HSPT-R7-008 - Answer A: One that remains filled with water far longer than intended

The passage states that a garden holding water too long may not function as intended.

121. HSPT-R7-009 - Answer D: It presents a runoff problem, explains one response, and discusses the response's limits and design needs.

The passage follows that problem-solution-and-qualification structure.

122. HSPT-R7-010 - Answer C: How quickly the soil drains

Soil drainage rate is explicitly listed as a design consideration.

123. HSPT-R7-011 - Answer B: Maps provide evidence that should be interpreted in light of the mapmaker's choices and purpose.

The passage emphasizes that maps contain useful evidence shaped by selection, audience, and purpose.

124. HSPT-R7-012 - Answer B: reliable and convincing.

In context, authoritative describes the impression that the map presents precise, dependable information.

125. HSPT-R7-013 - Answer C: Different users may need different information about the same region.

Merchant and military maps emphasize different features because their users have different priorities.

126. HSPT-R7-014 - Answer D: To illustrate how a map's purpose affects what it emphasizes

The contrast demonstrates that different map purposes lead to different choices about prominent information.

127. HSPT-R7-015 - Answer B: analytical and measured

The author examines maps critically without dismissing their usefulness.

128. HSPT-R7-016 - Answer C: letters and surveys

The passage explicitly says historians may compare a map with letters, surveys, or later maps.

129. HSPT-R7-017 - Answer D: She assumes she may have damaged it.

Mara apologizes because she believes the tear may have occurred while she was handling the lantern.

130. HSPT-R7-018 - Answer C: He values objects that can still serve a purpose.

He kept the lantern because its frame remained usable and then helped repair its torn paper.

131. HSPT-R7-019 - Answer B: It shows that an imperfection can be viewed creatively.

The child sees the visible patch not as a flaw but as an interesting mountain-like shape.

132. HSPT-R7-020 - Answer C: Something can remain worthwhile even when it is imperfect.

Mara's experience with the repaired lantern changes her belief that usefulness requires a flawless condition.

133. HSPT-R7-021 - Answer D: placed so that it matched the space.

They positioned the translucent paper behind the tear so it properly covered the damaged area.

134. HSPT-R7-022 - Answer B: They can move toward the beetle's mouth.

The passage states that larger combined droplets can move toward the beetle's mouth with help from gravity.

135. HSPT-R7-023 - Answer D: To explain a beetle's water-collection process and how it can inspire engineering research

The passage explains the biological process and then describes how researchers study its principles for possible human applications.

136. HSPT-R7-024 - Answer D: Engineered collectors must satisfy conditions beyond the beetle's natural process. Human designs must account for factors such as cost, durability, humidity, wind, and scale.
137. HSPT-R7-025 - Answer C: It explains a natural process, then discusses a possible application and its limits. The passage moves from beetle behavior to engineering interest and then qualifies the challenges of applying the idea.
138. HSPT-R7-026 - Answer B: Local humidity
The final paragraph explicitly lists local humidity among the factors a human-made collector must account for.
139. HSPT-R7-027 - Answer B: Distinguish known information from uncertainty.
The passage says a careful label separates what is known from what remains uncertain.
140. HSPT-R7-028 - Answer A: Museum labels result from research and choices about accuracy, emphasis, and audience. The passage repeatedly emphasizes evidence, selection, uncertainty, visitor needs, and revision.
141. HSPT-R7-029 - Answer D: They may judge that it distracts from the most useful idea for visitors.
The passage explains that too many technical details can obscure the central idea a label is meant to communicate.
142. HSPT-R7-030 - Answer A: To emphasize that a short label reflects substantial work and decision-making
The phrase contrasts the label's small visible form with the research and choices behind it.
143. HSPT-R7-031 - Answer C: It describes stages and considerations involved in creating and updating labels.
The passage moves through research, writing for visitors, revision, and the broader meaning of those choices.
144. HSPT-R7-032 - Answer C: To show that changing a label can reflect the growth of knowledge
The author explains that new evidence can improve interpretations and that revision can demonstrate developing historical knowledge.
145. HSPT-R7-033 - Answer A: a fallen branch
The passage states that a fallen branch hid one of the wooden trail markers.
146. HSPT-R7-034 - Answer D: He remembers map details that match the surrounding land.
Eli connects the map's creek bend with the visible slope and groove, giving evidence for the narrower route.
147. HSPT-R7-035 - Answer B: Evidence can be especially valuable when familiar conditions change.
The changed trail causes Soren's familiarity to be less reliable, while Eli's attention to evidence guides them correctly.
148. HSPT-R7-036 - Answer D: likely to be correct
The wider path looks more believable as the correct route, so convincing means seeming likely to be right.
149. HSPT-R7-037 - Answer A: Information that can become difficult to notice when circumstances change
The marker still exists but is obscured, reinforcing the story's focus on checking evidence when familiar cues become unreliable.
150. HSPT-R7-038 - Answer A: Its painted surface could scrape against the floor.
The passage directly explains that they move slowly to prevent the painted panel from scraping the floor.
151. HSPT-R7-039 - Answer D: She connects the careful pace of carrying the panel with the pace of the music.
The deliberate movement remains in Leena's mind and helps her allow each musical phrase to finish.
152. HSPT-R7-040 - Answer D: An insight from one activity can change how a person approaches another.
Leena improves not through more rapid repetition but by applying an awareness of pace learned during a different task.
153. HSPT-R7-041 - Answer A: To show that Leena feels pressure even without an actual audience
Leena imagines the empty seats judging her, revealing that her nervousness comes from her own anticipation.
154. HSPT-R7-042 - Answer B: A problem is introduced, an unrelated-seeming event occurs, and that event helps produce a solution.
Leena struggles, helps with the panel, and then uses the sense of deliberate pace to improve her playing.
155. HSPT-R7-043 - Answer D: Cleaning items between loans
The passage lists cleaning among the practical issues staff must consider when lending objects.
156. HSPT-R7-044 - Answer C: Libraries of things apply the principle of shared access to useful objects while requiring careful planning.
The passage explains both the shared-access principle and the operational decisions needed to make these collections useful.
157. HSPT-R7-045 - Answer C: Community needs and requests can differ by location.
The passage says libraries study local needs and adjust inventory, implying different communities may benefit from different objects.
158. HSPT-R7-046 - Answer D: the collection of available items
In context, inventory refers to the set of objects the library keeps available for lending.

159. HSPT-R7-047 - Answer C: It defines a practice, explains benefits and challenges, then describes how local needs shape it.
The passage introduces libraries of things, discusses reasons and operational concerns, and then explains community-based selection.
160. HSPT-R7-048 - Answer C: It was taken in another town.
The caption reveals that the photograph was taken elsewhere.
161. HSPT-R7-049 - Answer A: She learns that it records an early stage in a local institution's history.
The note identifies the temporary library, giving the image important historical context.
162. HSPT-R7-050 - Answer A: Nora's opportunity to decide what kind of history deserves attention
The empty frame creates the choice that leads Nora to value an understated image.
163. HSPT-R7-051 - Answer B: professionally impressive in appearance
The faded image is contrasted with more visually refined photographs.
164. HSPT-R7-052 - Answer A: Ordinary-looking evidence can reveal important parts of a community's story.
Nora learns that a modest photograph can reveal meaningful community history.
165. HSPT-R7-053 - Answer B: It slows down.
The passage states that wind slows on the downwind side.
166. HSPT-R7-054 - Answer A: The fence may cause a drift where the road needs to remain clear.
Poor placement can cause snow to settle in an undesirable location.
167. HSPT-R7-055 - Answer B: opposite to what one might first expect
The surprising principle differs from the obvious idea of simply blocking snow.
168. HSPT-R7-056 - Answer A: A mechanism, placement factors, examples, and a concluding principle
The passage moves through how snow fences work, design factors, examples, and the central principle.
169. HSPT-R7-057 - Answer B: To explain how snow fences use changes in wind speed to control where drifting snow settles
The passage explains the physical principle and placement choices behind snow fences.
170. HSPT-R7-058 - Answer A: A recently changed street name
The passage specifically lists a recently changed lane name.
171. HSPT-R7-059 - Answer C: The notes reveal uncertainties and decisions hidden by the map's finished appearance.
The notes expose questions, disagreements, and judgments behind the final map.
172. HSPT-R7-060 - Answer C: resolved or decided
The printed map presents choices as already resolved.
173. HSPT-R7-061 - Answer D: Clean final maps versus working notes that preserve uncertainty
The passage repeatedly contrasts finished representation with notes revealing unresolved questions.
174. HSPT-R7-062 - Answer A: Working notes can reveal the judgments and uncertainties behind a finished map.
The passage focuses on reasoning and uncertainty preserved in mapmakers' notes.

Mathematics

175. HSPT-M7-001 - Answer A: 11

Multiply first: $3 \times 4 = 12$; then $18 - 12 + 5 = 11$.

176. HSPT-M7-002 - Answer B: $9/20$

$0.45 = 45/100$, which simplifies to $9/20$.

177. HSPT-M7-003 - Answer D: 34

$3^3 = 27$ and $\sqrt{49} = 7$, so $27 + 7 = 34$.

178. HSPT-M7-004 - Answer A: 49

The positive factors of 49 are 1, 7, and 49, so it has exactly three.

179. HSPT-M7-005 - Answer D: 5

Divide by 3 to get $x + 4 = 9$, then subtract 4 to get $x = 5$.

180. HSPT-M7-006 - Answer C: $2n - 6$

Twice n is $2n$, and six less than that is $2n - 6$.

181. HSPT-M7-007 - Answer A: $F = 4 + 2.50m$

The fare combines the fixed \$4 charge with \$2.50 multiplied by the number of miles.

182. HSPT-M7-008 - Answer A: 8

Solve $5x - 8 = 32$: add 8 to get 40, then divide by 5 to get 8.

183. HSPT-M7-009 - Answer D: 75°

Quadrilateral angles total 360° ; $360 - (88 + 92 + 105) = 75^\circ$.

184. HSPT-M7-010 - Answer D: 94 cm^2

Surface area is $2(5 \times 3 + 5 \times 4 + 3 \times 4) = 2(47) = 94 \text{ cm}^2$.

185. HSPT-M7-011 - Answer B: 10 cm

Since $C = \pi d$, $d = 31.4 \div 3.14 = 10 \text{ cm}$.

186. HSPT-M7-012 - Answer B: 2,400

One kilometer equals 1,000 meters, so 2.4 kilometers equals 2,400 meters.

187. HSPT-M7-013 - Answer A: 27 feet

At 6 feet per inch, 4.5 inches represents $4.5 \times 6 = 27$ feet.

188. HSPT-M7-014 - Answer B: 18

The ordered middle value of the five numbers is 18.

189. HSPT-M7-015 - Answer A: $3/4$

Six of eight sections are not purple, so the probability is $6/8 = 3/4$.

190. HSPT-M7-016 - Answer B: 20

Four scores with mean 18 total 72; $72 - (12 + 19 + 21) = 20$.

191. HSPT-M7-017 - Answer D: 24

The smallest positive number divisible by both 8 and 12 is 24.

192. HSPT-M7-018 - Answer B: $5/6$

$5/6$ is about 0.833, which is greater than 0.82, 0.81, and 0.80.

193. HSPT-M7-019 - Answer C: $4 \frac{3}{8}$

$2 \frac{6}{8} + 1 \frac{5}{8} = 3 \frac{11}{8} = 4 \frac{3}{8}$.

194. HSPT-M7-020 - Answer B: 19

Substitute 7 for n : $3(7) - 2 = 21 - 2 = 19$.

195. HSPT-M7-021 - Answer A: 9

Expand to $2x - 6 + 5 = 17$, so $2x - 1 = 17$, $2x = 18$, and $x = 9$.

196. HSPT-M7-022 - Answer C: $x < 4$

Subtract 4 to get $5x < 20$, then divide by 5 to obtain $x < 4$.

197. HSPT-M7-023 - Answer D: 48

The rate is $18 \div 3 = 6$ bottles per minute; in 8 minutes it fills $6 \times 8 = 48$ bottles.

198. HSPT-M7-024 - Answer B: 10 cm

By the Pythagorean relationship, $6^2 + 8^2 = 36 + 64 = 100$, and $\sqrt{100} = 10$.

199. HSPT-M7-025 - Answer A: 77 cm²
The area of a parallelogram is base $\times$ height, so $11 \times 7 = 77$ cm².
200. HSPT-M7-026 - Answer A: 15 cm
The scale factor is $10/4 = 2.5$, so the corresponding side is $6 \times 2.5 = 15$ cm.
201. HSPT-M7-027 - Answer D: 3,500
One liter equals 1,000 milliliters, so 3.5 liters equals 3,500 milliliters.
202. HSPT-M7-028 - Answer A: 30 liters
The volume is $40 \times 25 \times 30 = 30,000$ cubic centimeters, which equals 30 liters.
203. HSPT-M7-029 - Answer B: \$0.30
Divide the total cost by the number of ounces: $\$3.60 \div 12 = \0.30 per ounce.
204. HSPT-M7-030 - Answer C: 12
The range is greatest minus least: $18 - 6 = 12$.
205. HSPT-M7-031 - Answer C: 7/12
There are 7 favorable tiles out of 12 total, so the probability is 7/12.
206. HSPT-M7-032 - Answer D: 90
A mean of 86 for five quizzes requires 430 points; the first four total 340, so the fifth score must be 90.
207. HSPT-M7-033 - Answer B: 31
The absolute values are 23 and 8, and $23 + 8 = 31$.
208. HSPT-M7-034 - Answer C: 6
First evaluate the parentheses, then multiply: $18 - 2 \times 6 = 18 - 12 = 6$.
209. HSPT-M7-035 - Answer D: 31/36
Using denominator 36 gives $21/36 + 10/36 = 31/36$.
210. HSPT-M7-036 - Answer C: \$58.80
A 30% discount leaves 70% of the price; $0.70 \times \$84 = \58.80 .
211. HSPT-M7-037 - Answer B: $2a + 10$
Combine like terms: $4a - 2a = 2a$ and $3 + 7 = 10$.
212. HSPT-M7-038 - Answer C: The initial charge before mileage is added
In $C = 4 + 2.5m$, the constant 4 is charged regardless of the number of miles.
213. HSPT-M7-039 - Answer B: 25
Let the integers be $n - 1$, n , and $n + 1$; then $3n = 72$, so $n = 24$ and the largest is 25.
214. HSPT-M7-040 - Answer C: $2n - 5$
Twice n is $2n$, and five less than that quantity is $2n - 5$.
215. HSPT-M7-041 - Answer D: 44 cm
Circumference is πd , so $(22/7) \times 14 = 44$ cm.
216. HSPT-M7-042 - Answer B: 75°
Quadrilateral angles total 360°, so $x = 360 - (90 + 85 + 110) = 75^\circ$.
217. HSPT-M7-043 - Answer C: 108 cm²
Surface area is $2(lw + lh + wh) = 2(24 + 18 + 12) = 108$ cm².
218. HSPT-M7-044 - Answer C: 63 miles
Multiply the scale by the map distance: $18 \times 3.5 = 63$ miles.
219. HSPT-M7-045 - Answer A: 6
Nine feet equals 108 inches, and $108 \div 18 = 6$ complete pieces.
220. HSPT-M7-046 - Answer D: 24
The data are already ordered, and the middle of five values is 24.
221. HSPT-M7-047 - Answer B: 1/8
The probabilities are 4/8 and 2/8; multiplying independent events gives $1/2 \times 1/4 = 1/8$.
222. HSPT-M7-048 - Answer D: The sets have the same mean, but Set B has a greater range.
Both sets total 30, so each mean is 6; their ranges are 6 and 11 respectively.
223. HSPT-M7-049 - Answer A: 5/8
 $5/8 = 0.625$, greater than 0.62, 0.61, and 0.6.
224. HSPT-M7-050 - Answer C: 42
The least positive number divisible by both 14 and 21 is 42.

225. HSPT-M7-051 - Answer A: 4.66
Numbers from 4.65 through 4.74 round to 4.7.
226. HSPT-M7-052 - Answer C: -10
Substitute -2: $3(-2) - 4 = -10$.
227. HSPT-M7-053 - Answer C: 5
Expand and solve: $3x + 6 = 2x + 11$, so $x = 5$.
228. HSPT-M7-054 - Answer D: $4 \leq x < 9$
At least 4 includes 4, while less than 9 excludes 9.
229. HSPT-M7-055 - Answer C: $18 + 4t$
The tank starts at 18 and gains 4 gallons for each minute t .
230. HSPT-M7-056 - Answer A: 15 cm
 $\sqrt{9^2 + 12^2} = \sqrt{225} = 15$ cm.
231. HSPT-M7-057 - Answer A: 9 cm
A regular hexagon has six equal sides, so $54 \div 6 = 9$ cm.
232. HSPT-M7-058 - Answer D: 20 cm
The scale factor is $15/6 = 2.5$; $8 \times 2.5 = 20$ cm.
233. HSPT-M7-059 - Answer D: 2 hours 25 minutes
The elapsed time is 2 hours 25 minutes.
234. HSPT-M7-060 - Answer A: 10 cups
 $2.5 \times 4 = 10$ cups.
235. HSPT-M7-061 - Answer D: 12 mph
 $42 \div 3.5 = 12$ miles per hour.
236. HSPT-M7-062 - Answer B: $7/10$
There are 7 non-blue tokens among 10 total.
237. HSPT-M7-063 - Answer B: 83
The total must be 328; subtracting 245 gives 83.
238. HSPT-M7-064 - Answer C: $3/10$
There are 60 votes total, and $18/60 = 3/10$.

Language

239. HSPT-L7-001 - Answer D: The box contained three items: a compass, a flashlight, and a whistle.
A colon correctly introduces a list after the complete clause "The box contained three items."
240. HSPT-L7-002 - Answer B: "I found the missing folder," Elena said.
A comma belongs inside the closing quotation mark before the dialogue tag.
241. HSPT-L7-003 - Answer D: The rain stopped; the players returned to the field.
A semicolon correctly joins two closely related independent clauses.
242. HSPT-L7-004 - Answer A: Our class begins Spanish lessons in September.
The language name Spanish and month September are proper names and are capitalized.
243. HSPT-L7-005 - Answer B: The club will meet on Memorial Day.
Both words in the official holiday name Memorial Day are capitalized.
244. HSPT-L7-006 - Answer D: embarrass
Embarrass is spelled with double r and double s.
245. HSPT-L7-007 - Answer C: The scientist recorded the occurrence in her notes.
Occurence is misspelled; the correct spelling is occurrence.
246. HSPT-L7-008 - Answer B: Each of the volunteers brings a reusable bottle.
The singular subject each requires the singular verb brings.
247. HSPT-L7-009 - Answer A: By the time we arrived, the movie had begun.
Had begun is the correct past perfect form for an action completed before another past action.
248. HSPT-L7-010 - Answer C: Walking through the station, Lena heard an announcement.
The introductory modifier correctly describes Lena, the person who was walking.
249. HSPT-L7-011 - Answer A: The new route is shorter than the old one.
Than is the correct conjunction for a comparison.
250. HSPT-L7-012 - Answer D: Maya told Priya that Priya's notebook was on the table.
Repeating Priya removes ambiguity about whose notebook is being discussed.
251. HSPT-L7-013 - Answer B: A simple exchange system helped students reuse school materials.
This sentence introduces the passage's central topic and previews the purpose of the swap shelf.
252. HSPT-L7-014 - Answer C: These changes made the shelf easier for students to use responsibly.
The sentence links the council's improvements with the successful student use described next.
253. HSPT-L7-015 - Answer B: Some students had notebooks with blank pages remaining, while others had binders, folders, or art supplies they no longer needed.
This revision combines both ideas clearly without changing their meaning.
254. HSPT-L7-016 - Answer C: The two cyclists' helmets were stored inside.
For a plural noun ending in s, the possessive apostrophe follows the final s: cyclists'.
255. HSPT-L7-017 - Answer C: After the final bell rang, we hurried to the bus.
A comma correctly follows the introductory dependent clause "After the final bell rang."
256. HSPT-L7-018 - Answer B: My aunt moved from Ohio to the Pacific Northwest.
Ohio and Pacific Northwest are geographic proper names and require capitalization; the common noun aunt does not.
257. HSPT-L7-019 - Answer A: We read a poem by Professor Elena Ruiz in English class.
A title used directly before a name is capitalized, as are the person's name and the language-derived course name English.
258. HSPT-L7-020 - Answer B: entrepreneur
Entrepreneur is the standard spelling.
259. HSPT-L7-021 - Answer A: The speaker gave a persistant reminder.
Persistant is misspelled; the standard spelling is persistent.
260. HSPT-L7-022 - Answer A: The principal made an immediate announcement.
Immediate and announcement are spelled correctly; the other choices contain noticeable, fertilizer, or rhythm misspelled.
261. HSPT-L7-023 - Answer B: The coach gave the forms to Jordan and me.
The pronoun is the object of the preposition to, so the objective form me is required.
262. HSPT-L7-024 - Answer D: The volunteers moved the supplies into the shelter.
This choice contains a subject and a complete predicate and does not depend on another clause.

263. HSPT-L7-025 - Answer C: The job requires sorting files, answering calls, and greeting visitors.
All three items use the same gerund form, creating parallel structure.
264. HSPT-L7-026 - Answer A: Please accept this gift as a token of thanks.
Accept means to receive willingly; except means to exclude.
265. HSPT-L7-027 - Answer C: The meeting was postponed until later.
This version communicates the idea clearly without redundant phrases such as "later time in the future."
266. HSPT-L7-028 - Answer A: Previously, the photographs were kept in a cabinet and used mainly during club meetings.
This revision preserves the meaning while eliminating unnecessary wording and maintaining a clear sentence structure.
267. HSPT-L7-029 - Answer B: These captions helped viewers connect each image with a specific period of local change.
The sentence directly explains the purpose of the captions and strengthens the transition to the group's decision about focus.
268. HSPT-L7-030 - Answer D: It introduces an idea the group evaluates and rejects to preserve the display's focus.
Sentence 5 sets up the decision in sentence 6, showing how the group maintained paragraph and display unity.
269. HSPT-L7-031 - Answer B: The supply box contained three items: tape, scissors, and string.
A colon correctly follows a complete clause that introduces a list.
270. HSPT-L7-032 - Answer B: Nora finished the poster; however, she forgot to add the date.
A semicolon separates the independent clauses, and a comma follows the conjunctive adverb however.
271. HSPT-L7-033 - Answer C: "Please close the window," said Mr. Lane.
In standard US dialogue punctuation, the comma stays inside the closing quotation mark before the dialogue tag.
272. HSPT-L7-034 - Answer C: Our class visits the National Air and Space Museum in October.
The official museum name and the month October are capitalized; ordinary words such as class are not.
273. HSPT-L7-035 - Answer D: On Memorial Day, the scouts met near Lake Crescent.
The holiday Memorial Day and the geographic proper name Lake Crescent require capitalization.
274. HSPT-L7-036 - Answer D: accommodate
Accommodate is correctly spelled with two c's and two m's.
275. HSPT-L7-037 - Answer C: The audience showed genuine enthusiasm.
Audience, genuine, and enthusiasm are all spelled correctly in this sentence.
276. HSPT-L7-038 - Answer A: Neither of the two routes leads to the station.
Neither is singular here, so it takes the singular verb leads.
277. HSPT-L7-039 - Answer D: Yesterday, Mia packed her bag and walked to the bus stop.
Both actions occurred yesterday, so the simple past forms packed and walked are consistent.
278. HSPT-L7-040 - Answer A: Maya photographed the branches covered in frost.
This placement clearly indicates that the branches, not Maya or the camera, are covered in frost.
279. HSPT-L7-041 - Answer C: The new rule may affect how teams schedule practice.
Affect is the verb meaning to influence, which is required after the modal verb may.
280. HSPT-L7-042 - Answer B: The scientist examined the sample.
Examined precisely describes careful observation, making it more specific than looked at.
281. HSPT-L7-043 - Answer D: However
However signals the contrast between the first route's shorter distance and the second route's different advantage.
282. HSPT-L7-044 - Answer C: The labels helped students choose seeds suited to the plants they wanted to grow.
This sentence explains the usefulness of the growing notes and develops the paragraph's focus on organizing the exchange.
283. HSPT-L7-045 - Answer D: By the end of the period, students had traded dozens of packets and left suggestions, which the club used to plan a second exchange for spring.
This revision combines the related ideas clearly while preserving the sequence from student feedback to future planning.
284. HSPT-L7-046 - Answer C: After the final bell rang, we hurried to the bus.
A comma follows the introductory dependent clause.
285. HSPT-L7-047 - Answer B: The two coaches' offices are beside the gym.
The plural possessive coaches takes an apostrophe after the final s.
286. HSPT-L7-048 - Answer A: My aunt studies Spanish and history at Western State University.
Spanish and the university's proper name are capitalized; history is lowercase.
287. HSPT-L7-049 - Answer A: Professor Elena Ruiz will speak at the City Arts Council meeting.
The title before the name and the organization's proper name are capitalized.

288. HSPT-L7-050 - Answer B: privilege

Privilege is the standard spelling.

289. HSPT-L7-051 - Answer A: maintenance

Maintenance is the standard spelling.

290. HSPT-L7-052 - Answer A: The committee reached a unanimous decision.

Committee, unanimous, and decision are all spelled correctly.

291. HSPT-L7-053 - Answer D: The coach gave Maya and me the schedule.

The pronoun is the object of gave, so me is required.

292. HSPT-L7-054 - Answer D: The volunteers returned to the field after the storm ended.

This choice contains an independent clause and expresses a complete thought.

293. HSPT-L7-055 - Answer C: The job requires sorting files, answering calls, and greeting visitors.

All three items use matching gerund forms.

294. HSPT-L7-056 - Answer B: The team is ready to begin.

This states the complete idea without redundant time expressions.

295. HSPT-L7-057 - Answer A: Lena, who was worried, spoke to Ava.

The relative clause clearly identifies Lena as worried.

296. HSPT-L7-058 - Answer D: As a result

As a result shows that the first outcome caused the revision.

297. HSPT-L7-059 - Answer D: Students often have strong opinions about noise, but the committee chose to investigate the issue with recorded evidence.

This introduces the noise issue and the committee's evidence-based approach.

298. HSPT-L7-060 - Answer C: The committee recommended repeating the survey on several different days so it could determine whether the same patterns appeared consistently before suggesting changes.

The revision clearly connects repeated measurement with checking consistency before recommendations.